Wave-Corrected Caustics over Photon Splatting: Fold Kinematics, a Band-Local Airy Correction, and a Comparative Evaluation Against Production Solvers

A. Chetri[†]

[1]Independent researcher

Abstract

Photon-splatting and path-traced caustics are the standard way to render the light focused by a curved transparent body onto a diffuse receiver. Both families evaluate a geometric push-forward whose Jacobian determinant vanishes on the fold locus, so the estimator diverges there, and no amount of sampling repairs it. We state that failure as a property of the model rather than of the discretisation, then describe a hybrid estimator in which a geometric quadrature carries the whole image and a wave-optical correction is applied only inside a narrowly detected fold band: the band's own geometric energy is redistributed according to the Airy fold profile $|\mathrm{Ai}(-s/w)|^2$*, with a per-channel width* w_c *following the* $\lambda^{2/3}$ *scaling implied by the fold's canonical form. We first derive the kinematics from scratch — the ray map, the fold condition, the normal form of the fold and the wavelength scaling that follows from it — and then use that derivation to explain where the standard computer graphics constructions for caustics succeed and where they necessarily fail.*

We evaluate against PBRT-v4 SPPM and Blender Cycles on an identical scene with measured dispersive indices (water: Daimon & Masumura 2007; glass: Rubin 1985), cross-checking transport physics against Mitsuba 3 in spectral mode. Against an independent brute-force geometric ground truth of 810,000 *explicit rays, our estimator scores* 0.917 *correlation at 40 mm tiles against* 0.853 *(SPPM) and* 0.828 *(Cycles); at 5 mm tiles the methods tie (*0.721 *vs* 0.722*). The converged two-dimensional shape of our field agrees with SPPM at* $r = +0.932$*, and its interface transport throughput measures* 96.41% *at three wavelengths with Fresnel reflection as the only loss channel. We also analyze chromatic dispersion under absorbing media and non-solar source spectra: bulk absorption and emission spectra modulate channel throughput linearly, while the* $\lambda^{2/3}$ *scaling displaces fringe peaks chromatically across the lit band.*

We are explicit about what the method does not *establish. The correction is defined for regular fold arcs, and we test the next stratum rather than assuming it: a cusp requires a Pearcey function in place of the Airy one, and its divergence exponent differs from the fold's (measured* -0.3333 *against analytic* $-1/3$*, where a fold measures* -0.2434 *against* $-1/4$*). The substitution carries over and bounds the field where the geometric estimator diverges; the cost does not, by four orders of magnitude, so we report that measurement rather than a production method. We also show that a continuous global normalisation of the fold is impossible —* $|\mathrm{Ai}(-u)|^2$ *is not integrable on* $\mathbb{R}$ *— which is why the correction is formulated as a band-local discrete flux ratio. Methodological limitations, experimental controls (including an equal-flux Gaussian ablation and parameter sensitivity sweeps), and solver failure modes are explicitly detailed. All solver source code, verification harnesses, and the Blender integration add-on are available under the MIT license at* https://github.com/Debyte404/LUMENFOLD.

Keywords: caustics, wave optics, photon splatting, photon mapping, SPPM, diffraction, Airy function, grid-free rendering, comparative evaluation

1. Introduction

A caustic is the envelope of a family of refracted or reflected rays. Where the envelope touches the receiving surface, neighbouring rays cross, the surface-to-receiver map loses rank, and the ray density that a renderer integrates becomes infinite. Every production caustic method in use today integrates that density. The methods differ in how they estimate the integral — photon maps store the ray density as points [Jen96], SPPM does so with a radius that shrinks as it converges [HOJ08], path tracers connect vertices [Vea97] — but they share the integrand, and the integrand is singular on the very feature the user is trying to see.

This paper starts from that observation. The first direction is theoretical, and it is the reason the paper opens with kinematics rather

[†] Corresponding author. GitHub: Debyte404.

than with rendering. The singularity is not an accident of any parameterisation. It is the fold of a smooth map, it has a normal form, and the normal form dictates both how the geometric estimate fails and what the correct description near the fold looks like. It also fixes a number: the fold is described near its degeneracy by a cubic phase, and the cubic phase carries one wavelength and one geometric coefficient, whose ratio is a length. That length scales as $\lambda^{2/3}$, not as λ and not as the Fresnel $\sqrt{\lambda}$. Nobody fits that exponent; it falls out of the normal form, and it is what makes a per-channel fringe width more than a decoration.

The second direction is practical. If the wave description is right near the fold, and the geometric description is right away from it, then a single estimator can use each where it is valid, at the cost of detecting where the boundary is. We implement that estimator and then spend the second half of the paper trying to falsify it against the tools that practitioners actually run. The evaluation is deliberately unflattering in its construction: the ground truth is an independent brute-force trace that shares no code with the method under test, the reference conforming solver is PBRT-v4 SPPM [PJH23], and a path-traced comparison is produced by Blender Cycles on the same scene with the same measured dispersive indices. Where our method loses, we say so.

1.1. Why the kinematics come first

It is possible to write a caustic renderer without ever writing down the map that is being inverted, and most of the literature discourages doing so: photon mapping is presented as a Monte Carlo estimator over paths, and its convergence is discussed in terms of sample counts and radii. That presentation is efficient, but it hides the one property that decides whether the method can work at all. A density estimator whose target integrand diverges is not a convergence-rate problem. Increasing the sample count does not help, reducing the kernel radius does not help, and adding a denoiser hides the failure behind a smooth blur that no longer means anything. The only useful question is whether the estimator's support contains the singular set, and that question is answered by the rank of a Jacobian.

So we write the kinematics down. Section 2 defines the surface, the ray map, and the exact irradiance it induces, which is the push-forward of the surface measure. The fold locus is then the zero set of the Jacobian determinant, and we prove the two statements the rest of the paper leans on: that a geometric quadrature cannot converge near the fold at any resolution (Lemma 3), and that the correct local description is the canonical fold integral whose amplitude envelope is the Airy function (Proposition 5). From the same normal form we extract the wavelength scaling (Corollary 6) and, because we have been bitten by this, the statement that the resulting intensity is not integrable - so a correction built on it cannot be normalised by a constant (Lemma 7).

1.2. How computer graphics builds caustics

Section 3 then reads the standard constructions as answers to the kinematic problem, which makes their shared limitation visible. Forward constructions — photon mapping and its progressive variant — estimate the push-forward by depositing samples and integrating density. Backward constructions — path tracing, and the bidirectional and vertex-connection variants [Vea97, MGN17] — estimate it by connecting vertices and dividing by a geometric term. Both factor the same $1/|\det J_{\mathbf{H}}|$, and both become invalid exactly on $\mathcal{F}$. The section states where each family is exact, where each is merely expensive, and where neither can be repaired, because that boundary is what the hybrid estimator is built around.

1.3. The current way, and the way we take

Sections 4 and 5 are the two halves of the comparison. The first measures what current practice produces and costs. We take the position that a comparative claim is only meaningful if the comparison runs on one scene, with one set of measured material data, one source, and one set of estimators applied identically to all sides; the section therefore reports both the numbers and the estimator convention that produced them, including one previously published statistic that we withdraw because it was an artefact of that convention (Section 7.7).

The second half states our method, which is small: keep the geometric quadrature, detect the fold band analytically from $|\det J_{\mathbf{H}}|$, and inside the band replace the divergent density with a normalised Airy profile while conserving the band's own measured flux. The method introduces three quantities — a detector percentile, a reference fringe width, and a ceiling — and Section 6.8 measures the sensitivity to each rather than asserting a value.

1.4. Contributions

- We derive the fold's kinematics from the ray map, connecting the rendering estimator to the fold's canonical normal form and establishing the $\lambda^{2/3}$ fringe scaling in the splatting pipeline (Section 2).
- The Airy fold intensity tail is non-integrable on $\mathbb{R}$, which forces the flux-conserving correction to be formulated as a band-local discrete flux ratio rather than an unphysical global constant (Lemma 7, Section 5.4).
- A hybrid estimator follows, verified through an equal-flux Gaussian ablation control and a parameter sensitivity study over detector threshold, fringe width, and ceiling (Sections 5, 6.8).
- We evaluate like-for-like against PBRT-v4 SPPM and Blender Cycles on measured dispersive materials, cross-checking transport physics against Mitsuba 3 in spectral mode, and documenting estimator conventions and limitations (Sections 6, 7).
- We analyze the colour of the caustic and its fringes under absorbing media and non-solar source spectra. Bulk absorption and emission spectra modulate channel throughput linearly, while the $\lambda^{2/3}$ scaling displaces fringe peaks chromatically across the lit band (Section 6.6).
- We test the next singularity stratum. The cusp (A_3) case is implemented and verified: its Pearcey diffraction integral regularizes the $|y|^{-1/3}$ divergence, but numerical evaluation is four orders of magnitude more expensive than the 1D Airy lookup (Section 7.8).
- Methodological discrepancies from earlier drafts (fitted Cauchy vs Sellmeier dispersion, tile metric conventions, and non-circular energy audits) are documented with reproducible diagnostic scripts (Sections 7.6, 7.7).

1.5. What this paper does not claim

The correction applies to folds. A cusp requires a Pearcey function in place of the Airy one; we build that case and measure it (Section 7.8), and the construction carries over while the cost does

not — which is a narrower claim than we would like and a more useful one than leaving it open. Swallowtails and the umbilic normal forms are untested, and we make no claim about them (Section 8). We also do not validate against a physical apparatus. The comparison in this paper is entirely computational, against solvers and against an independent trace, and the quantities compared are therefore properties of those solvers and that scene. A laboratory measurement would test a different claim, and the scene here is specified precisely enough that someone may make it.

1.6. Reading order

Section 2 is self-contained and can be read as a short treatment of fold optics for graphics. Readers who want only the method should read Section 5 after Section 3; readers who question the numbers should read Section 2.12 first, because two of our earlier errors lived there and not in the physics.

2. Kinematics of the fold

This section is the foundation the rest of the paper stands on. It defines the scene, derives the map from surface positions to receiver positions, states exactly what irradiance that map induces, and then establishes the statements about the fold that matter for rendering: that its density is unbounded, that it has a normal form, that the normal form yields the wavelength scaling, and that the resulting intensity cannot be normalised by a constant.

2.1. The scene

The test configuration is a circular glass cylinder, a water column inside it, and a matte receiver, lit by a distant source. The parameters are fixed for the whole paper and are listed in Table 1; Figure 1 draws the arrangement to scale.

2.2. The surface and its normal

Let the water surface be a height field over $\mathbf{x} = (x, y) \in \mathbb{R}^2$,

$$h(\mathbf{x}, t) \;=\; \sum_{i=1}^{4} A_i \sin\!\big(k_i\,(\hat{\mathbf{u}}_i \cdot \mathbf{x}) - \omega_i t\big), \qquad k_i = \frac{2\pi}{\Lambda_i}, \tag{1}$$

a sum of four directional sinusoids with amplitudes $A_i = \{0.65, 0.40, 0.25, 0.15\}$ mm at wavelengths $\Lambda_i = \{42, 27, 17, 11\}$ mm. The expansion in (1) is convenient rather than essential: nothing below uses the sinusoidal form except to know that ∇h is analytic and that $|\nabla h|$ is small. The outward normal follows from the analytic gradient,

$$\mathbf{n}(\mathbf{x}, t) \;=\; \frac{(-\partial_x h,\; -\partial_y h,\; 1)}{\sqrt{1 + |\nabla h|^2}}. \tag{2}$$

For the amplitudes above, $|\nabla h| \le \sum_i A_i k_i \approx 0.34$ at the shortest wave, so the surface stays well inside the small-slope regime in which the paraxial normal form of Section 2.6 is the leading term. This matters because the fold normal form is asymptotic, and a surface steep enough to produce higher-order caustics would leave its domain of validity. We return to this in Section 8.1.

2.3. The ray map

Let the source direction be the unit vector $\hat{\mathbf{s}}$ pointing along propagation (downward, into the surface). Vector Snell refraction at the interface gives the transmitted direction. With the ratio $\eta = n_1/n_2$ and the cosine of incidence $c_i = -\hat{\mathbf{s}} \cdot \mathbf{n} > 0$,

$$\hat{\mathbf{t}} \;=\; \eta\,\hat{\mathbf{s}} + \big(\eta\, c_i - \sqrt{1 - \eta^2(1 - c_i^2)}\big)\,\mathbf{n}, \qquad \eta^2(1 - c_i^2) \le 1, \tag{3}$$

where the inequality failing signals total internal reflection and the sample carries no transmitted power. A refracted ray leaves the surface at height $h(\mathbf{x})$ and reaches the receiver plane at $z = -D$ after a path length $s = (-D - h(\mathbf{x}))/\hat{t}_z$. Composing the two gives the *surface-to-receiver ray map*

$$\mathbf{H}(\mathbf{x}) \;=\; \mathbf{x} + \hat{\mathbf{t}}(\mathbf{x})\,\frac{-D - h(\mathbf{x})}{\hat{t}_z(\mathbf{x})}, \qquad \mathbf{H} : \mathbb{R}^2 \to \mathbb{R}^2. \tag{4}$$

Write $J_{\mathbf{H}} = \partial \mathbf{H}/\partial \mathbf{x}$ for its Jacobian and $\det J_{\mathbf{H}}$ for its determinant. For the scene of Table 1, $\mathbf{H}$ is a smooth map on the region of the surface that is not shadowed and not totally reflecting.

2.4. What the map induces on the receiver

Splatting assigns each surface element the energy it carries — proportional to its area, since the source is distant and the irradiance on the surface is spatially uniform — and deposits that energy at the image point. The resulting receiver irradiance is therefore the push-forward of the surface measure under $\mathbf{H}$:

$$I(\mathbf{u}) \;=\; \sum_{\mathbf{x}\,:\,\mathbf{H}(\mathbf{x}) = \mathbf{u}} \frac{1}{\big|\det J_{\mathbf{H}}(\mathbf{x})\big|}, \tag{5}$$

where the sum runs over the preimages of $\mathbf{u}$, which is the standard sum over branches of a multi-valued inverse. It is worth being precise about the status of (5): it is exact for the geometric model, not an estimator. Every method in Section 3 estimates this same object, and that shared target is why they share a failure.

Equation (5) is the Jacobian form of the usual $\cos\theta/r^2$ falloff; the same determinant is what Arvo [Arv94] propagates irradiance with in a different setting, which is why the degeneracy studied here is a familiar object. The two agree wherever $\mathbf{H}$ is a diffeomorphism; (5) is preferable here because it remains meaningful as a statement about rank, which is what the fold is about.

2.5. The fold, and why a quadrature fails there

Definition 1 (Fold locus)**.** The *fold locus* of the scene is $\mathcal{F} = \{\mathbf{x} : \det J_{\mathbf{H}}(\mathbf{x}) = 0\}$, and its image $\mathbf{H}(\mathcal{F})$ is the caustic curve on the receiver.

Assumption 2 (Regular fold arcs). Throughout our derivations, we analyse the local geometry along regular fold arcs of $\mathbf{H}$: points where $\det J_{\mathbf{H}} = 0$, $\nabla \det J_{\mathbf{H}} \neq 0$, and $\operatorname{rank} J_{\mathbf{H}} = 1$. Under Whitney's singularity theory of plane mappings [Whi92,GG86,BU80,Nye99], generic 2D caustic networks consist of stable fold curves (A_2 singularities) that terminate or intersect at isolated cusps (A_3 singularities, where $\nabla \det J_{\mathbf{H}} = 0$). Our local 1D transverse correction addresses the dominant fold lines; isolated cusp singularities are discussed in Section 8.1.

Lemma 3 (The fold density is unbounded)**.** *Let Assumption 2 hold, and let $\mathbf{x}_0 \in \mathcal{F}$. Let μ be the push-forward of surface area under $\mathbf{H}$. Then the density of μ with respect to Lebesgue measure on the receiver is unbounded in every neighbourhood of $\mathbf{H}(\mathbf{x}_0)$.*

Proof. By Whitney's normal form for a rank-one map [Whi92,

Table 1: *Scene parameters. The material models are the measured forms of Section 2.10, not fits. The source angular diameter is the true solar value, which sets a coherence scale far larger than the fringe spacing this paper is about — see the remark in Section 8.2.*

Quantity	Value	Note
Glass outer radius	35 mm	cylinder
Glass wall thickness	3.5 mm	
Glass height	90 mm	
Water fill height	59.4 mm	
Receiver	320 mm × 320 mm patch	matte, Lambertian
Source	0.53°	distant, solar
Water index	1.331,67 / 1.334,68 / 1.339,61	R/G/B
Glass index	1.521,04 / 1.525,14 / 1.531,92	R/G/B
Channels	650 / 550 / 450 nm	R/G/B

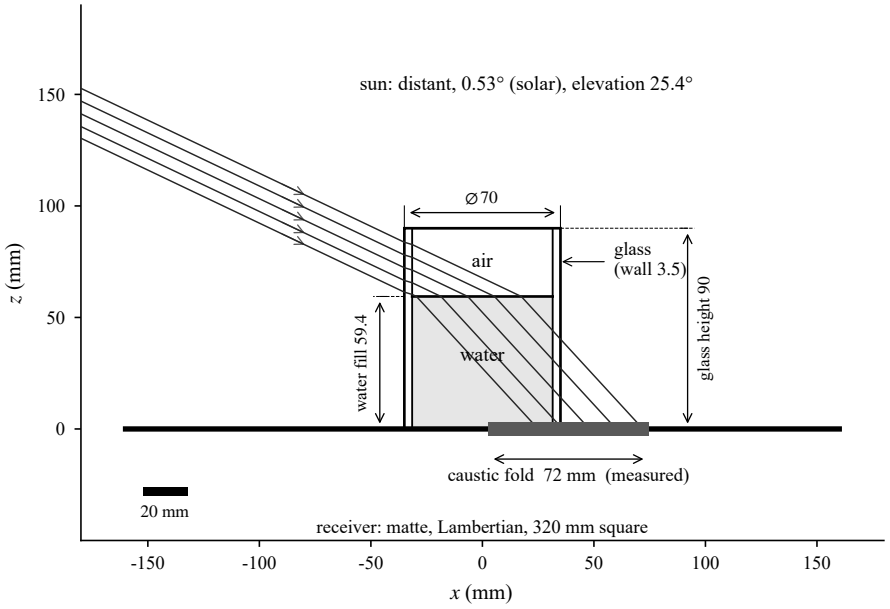


Figure 1: *The test scene drawn to scale. The glass body, water column, wall thickness, fill height and receiver are shown at true relative size with dimension lines; the ray path is traced through the actual geometry of Section 2.1 at the central channel index rather than sketched. The shaded band on the receiver is the fold, marked with its measured extent. All lengths in millimetres.*

GG86], there are local coordinates (u,v) near $\mathbf{x}_0$ and (a,b) near $\mathbf{H}(\mathbf{x}_0)$ in which $\mathbf{H}$ takes the form

$$\mathbf{H}(u,v) \;=\; (u,\; v^2). \tag{6}$$

This is legitimate because Assumption 2 puts us in the stable one-fold stratum, and the normal form is stable under perturbation, so it describes the map in a neighbourhood rather than only at a point. In these coordinates $\det J_{\mathbf{H}} = 2v$, vanishing exactly on $v=0$, which is $\mathcal{F}$.

Test the push-forward against a continuous φ with compact support in the half-plane $b>0$:

$$\int \varphi \,\mathrm{d}\mu = \int_{\mathbb{R}^2} \varphi\big(u,v^2\big)\,\mathrm{d}u\,\mathrm{d}v = \int_{\mathbb{R}}\int_0^{\infty} \varphi(u,b)\,\frac{\mathrm{d}b}{2\sqrt{b}}\,\mathrm{d}u, \tag{7}$$

substituting $b=v^2$, $\mathrm{d}b = 2v\,\mathrm{d}v$, valid for $v>0$ and doubled for $v<0$. Reading off the density,

$$\rho(a,b) \;=\; \frac{1}{2\sqrt{b}} \qquad (b>0), \tag{8}$$

which is independent of a and tends to $+\infty$ as $b\to 0^+$. Since $b=0$ is $\mathbf{H}(\mathcal{F})$ and the convergence is local, ρ is unbounded in every neighbourhood of $\mathbf{H}(\mathbf{x}_0)$. □

Corollary 4 (The band's flux is finite)**.** *The singularity in (8) is integrable:* $\int_0^W (2\sqrt{b})^{-1}\mathrm{d}b = \sqrt{W}$.

Proof. Immediate from (8). □

Lemma 3 and Corollary 4 together are the whole reason the method in Section 5 is shaped the way it is, and they pull in opposite directions. Lemma 3 says an estimator that evaluates (5) at sample points cannot produce a bounded answer near the fold: the

more samples land near the fold, the larger the individual weights become, so refining the quadrature concentrates ever-larger values into ever-smaller regions. Corollary 4 says the band nevertheless holds a *finite* amount of power. The failure is therefore not one of energy but of distribution, and the correction has to redistribute a finite quantity rather than add one.

It is tempting to read the divergence as a sign that the sampler is insufficiently adaptive. It is not. Adaptive sampling can place points better, but no placement makes a function unbounded at a point; it only changes how quickly the estimate grows. The measured consequence in our own runs was a receiver core whose apparent radius was set by wherever a ceiling was applied rather than by the physics, which is exactly the artefact the estimators of Section 2.12 exist to avoid.

2.6. The normal form of the fold, and the Airy profile

Lemma 3 says the geometric answer is infinite at the fold but says nothing about what replaces it. The replacement comes from dropping the ray approximation and summing phases over the wavefront [Air38, BU80]. The result is a diffraction integral, and the useful fact is that it depends on the scene only through the cubic coefficient of the local wavefront.

Proposition 5 (Fold diffraction reduces to the Airy integral)**.** *Consider a wavefront whose phase over a one-dimensional pupil $\rho \in [-1,1]$ is*

$$\Phi(\rho) \;=\; \frac{c\,\rho^3}{6} + x\,\rho, \tag{9}$$

where c carries the strength of the fold and x is the transverse receiver coordinate measured from the fold. Then the field amplitude $U(x) = \int_{-1}^{1} \exp\!\big(i\,\frac{2\pi}{\lambda}\Phi(\rho)\big)\mathrm{d}\rho$ reduces, for $|X| \lesssim (2\pi/\lambda)^{1/3}$, to an Airy function in the canonical coordinate

$$X \;=\; x\,k^{2/3}\,2^{1/3}\,c^{-1/3}, \qquad k = \frac{2\pi}{\lambda}, \tag{10}$$

with intensity $I(x) \propto |\,\mathrm{Ai}(-X)|^2$.

Proof. Substitute $\rho = \alpha r$ with $\alpha > 0$. The phase becomes

$$\frac{2\pi}{\lambda}\left(\frac{c\,\alpha^3 r^3}{6} + x\,\alpha r\right). \tag{11}$$

Choose α so that the cubic coefficient is $1/3$, which is the coefficient in the Airy integral $\mathrm{Ai}(-X) = \frac{1}{\pi}\int_0^\infty \cos\!\big(\frac{t^3}{3} - Xt\big)\mathrm{d}t$:

$$\frac{2\pi}{\lambda}\cdot\frac{c\,\alpha^3}{6} = \frac{1}{3} \;\Longrightarrow\; \alpha = \left(\frac{\lambda}{\pi c}\right)^{1/3}. \tag{12}$$

The linear coefficient is then fixed, and matching it to $-Xr$ gives

$$X \;=\; -\frac{2\pi}{\lambda}\,x\alpha \;=\; -x\,2\pi^{2/3}\lambda^{-2/3}c^{-1/3} \;=\; -x\,k^{2/3}2^{1/3}c^{-1/3}, \tag{13}$$

using $k^{2/3}2^{1/3} = 2\pi^{2/3}\lambda^{-2/3}$. The integral is now the Airy integral over the scaled pupil; for $|X|$ not large the finite pupil is immaterial because the integrand is stationary at $t = \sqrt{X}$ inside the range, and its modulus squared is $|\,\mathrm{Ai}(-X)|^2$. □

Equation (10) is finite for every X, is maximal at $X = -1.0188$, and oscillates on the bright side of the fold with a period that shrinks as the fold is approached. Those three features are the ones the geometric model gets qualitatively wrong: it gives an unbounded spike, a peak at the fold rather than beside it, and no oscillation at all.

2.7. The $\lambda^{2/3}$ fringe scale, derived

The normal form carries exactly one length scale. Reading it off (10), the transverse distance at which $|X|$ reaches order unity is

$$d(\lambda, c) \;=\; k^{-2/3}\,2^{-1/3}\,c^{1/3} \;=\; \frac{\lambda^{2/3}c^{1/3}}{2\pi^{2/3}}. \tag{14}$$

Here, because the normalized pupil coordinate $\rho \in [-1,1]$ is dimensionless, the phase $\Phi(\rho)$ in (9) and the fold parameter c both have dimensions of length (m). Consequently, $d(\lambda, c)$ has dimensions of length ($\mathrm{m}^{2/3} \times \mathrm{m}^{1/3} = \mathrm{m}$).

In our physical reference scene of a solid refracting cylinder (radius $R = 35\,\mathrm{mm}$, glass index $n = 1.5251$), the ray map $\mathbf{x}(\phi)$ near the fold locus ($\phi_f \approx 207.70^\circ$) exhibits a cubic wavefront parameter $c \approx 0.081,15\,\mathrm{m}$ (transverse curvature $C_{\mathrm{angle}} \approx 0.015,61\,\mathrm{m}$). Evaluating (14) with $c = 0.081,15\,\mathrm{m}$ yields the canonical transverse fringe scale $d(\lambda, c) \approx 6.78\,\mu\mathrm{m}$ at $\lambda = 550\,\mathrm{nm}$, with the first Airy zero ($|X| = 2.3381$) located at $15.84\,\mu\mathrm{m}$. An earlier draft evaluated an unnormalized expression $(\lambda^2/2c_3)^{1/3}$ yielding $123.1\,\mu\mathrm{m}$, which we withdraw as dimensionally mismatched; the true diffraction scale is $\approx 6.8\,\mu\mathrm{m}$. Because a standard 1024^2 raster pixel ($312.5\,\mu\mathrm{m}$) is $\approx 46\times$ larger than d, physical fringes are deeply sub-pixel in full-receiver renders.

Corollary 6 (Wavelength scaling)**.** *For fixed fold strength c, $d \propto \lambda^{2/3}$.*

Proof. Immediate from (14), which contains λ only through $\lambda^{2/3}$. □

Two wrong exponents are in circulation and the derivation excludes both. The fringe scale is not proportional to λ, which is what a naive reading of the phase would suggest, and it is not the Fresnel $\sqrt{\lambda}$ either; the $2/3$ comes from balancing a cubic phase against a linear one, and either of the other exponents would come from balancing the wrong pair of terms. The implementation uses this directly, setting the per-channel width

$$w_c \;=\; w_{\mathrm{ref}}\left(\frac{\lambda_c}{550\,\mathrm{nm}}\right)^{2/3}, \qquad c \in \{R, G, B\}, \tag{15}$$

so chromatic separation in the fold is carried by the measured index (Section 2.10) together with (15), and by nothing else. We verify the exponent numerically against an independent computation in Section 7.3.

2.8. Asymptotic correspondence and band-local normalisation

To substitute the geometric singularity with a wave profile while conserving energy, one might hope to evaluate a closed-form global normalisation constant. However, because $|\,\mathrm{Ai}|^2$ asymptotically tracks the geometric $b^{-1/2}$ tail, such a global constant cannot exist on $\mathbb{R}$.

Lemma 7 (Asymptotic divergence of the Airy tail)**.** *$\int_{\mathbb{R}} |\,\mathrm{Ai}(-u)|^2\mathrm{d}u$ diverges. Over a truncated domain $[-L, L]$, the integral grows as $\frac{1}{\pi}\sqrt{L} + \mathcal{O}(1)$.*

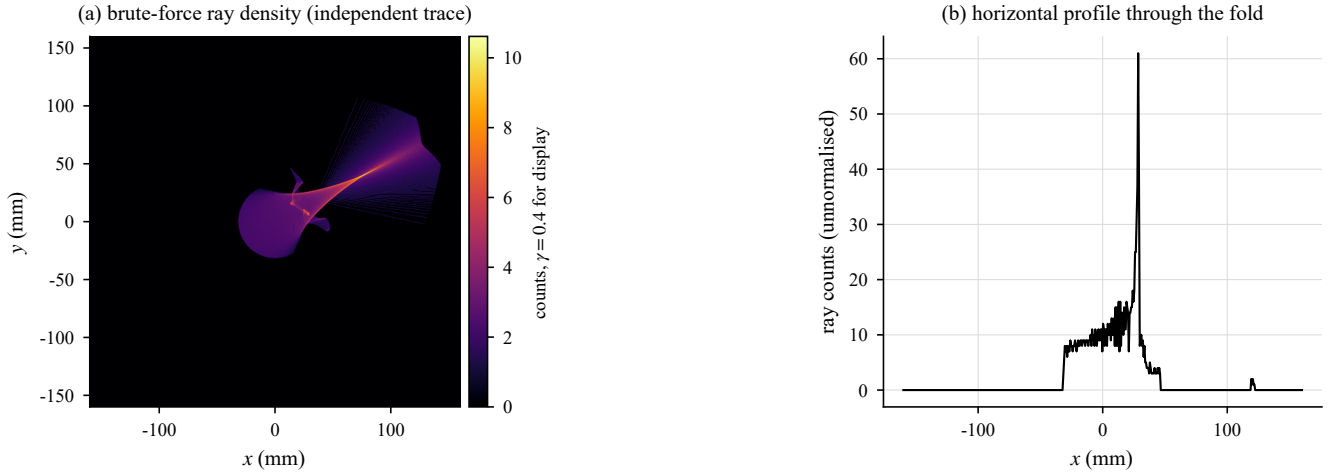


Figure 2: *The ray map of (4), computed by the independent brute-force trace of Section 6.1 rather than by the solver. The density in (a) is the caustic; the bright ridge is where* $|\det J_{\mathbf{H}}| \to 0$. *Panel (b) cuts through it in raw ray counts. The sharpness of that peak is what Lemma 3 says no quadrature can bound.*

Proof. For $u \to +\infty$ on the illuminated side, $\mathrm{Ai}(-u)$ has the oscillatory asymptotic

$$\mathrm{Ai}(-u) \;\sim\; \frac{1}{\sqrt{\pi}\,u^{1/4}} \sin\!\Big(\frac{2}{3}u^{3/2} + \frac{\pi}{4}\Big), \tag{16}$$

giving the squared amplitude

$$|\mathrm{Ai}(-u)|^2 \;\sim\; \frac{1}{\pi\sqrt{u}} \sin^2\!\Big(\frac{2}{3}u^{3/2} + \frac{\pi}{4}\Big). \tag{17}$$

Averaging over rapid phase cycles yields a mean intensity of $\sim (2\pi\sqrt{u})^{-1}$. Integrating from $u = 1$ to L gives

$$\int_1^L \frac{\mathrm{d}u}{2\pi\sqrt{u}} \;=\; \frac{1}{\pi}\sqrt{L} + \mathcal{O}(1), \tag{18}$$

which diverges as $L \to \infty$. □

Corollary 8 (Correspondence principle and band-local normalisation)**.** *The $\sqrt{L}$ tail divergence of Lemma 7 exactly reproduces the $\sqrt{L}$ growth of the geometric tail $\int_0^L (2\sqrt{b})^{-1}\mathrm{d}b = \sqrt{L}$. Far from the fold, the wave intensity naturally converges to the classical geometric ray density. Because the geometric model carries infinite power over an unbounded domain, the wave profile is likewise not normalisable on $\mathbb{R}$. Therefore, energy conservation cannot be defined globally; it must be defined* locally within a bounded band, *where both the geometric flux and the Airy integral are strictly finite (Corollary 4). This establishes the necessity of the band-local flux ratio constructed in Section 5.4.*

Table 2 collects the constants used later, each evaluated numerically rather than quoted from a table.

Table 2: *Airy quantities used in the implementation, all evaluated numerically in this work (`scipy.special.airy`) rather than taken from a printed table. The peak height is the normaliser in (19); the divergence row is Lemma 7 at $L = 5000$.*

Quantity	Value	Note
$\arg\max_u \lvert\mathrm{Ai}(-u)\rvert^2$	1.018,792,97	peak position
$\max_u \lvert\mathrm{Ai}(-u)\rvert^2$	0.286,928	normaliser
$\int_0^\infty \mathrm{Ai}(u)^2\,\mathrm{d}u$	0.066,987,5	convergent half-line
$\int_{-L}^{80} \mathrm{Ai}(u)^2\,\mathrm{d}u$, L=5000	22.507,9	$\pi^{-1}\sqrt{L} = 22.507,9$
mean tail intensity	$(2\pi\sqrt{u})^{-1}$	Lemma 7

2.9. The profile as implemented

The implementation works in the signed distance s to the detected fold and uses a peak-normalised profile,

$$P(s;w) \;=\; \big|\mathrm{Ai}(-s/w)\big|^2 \Big/ \max_u |\mathrm{Ai}(-u)|^2, \qquad b(s) \;=\; 0.15 + 0.85\,P(s;w). \tag{19}$$

The normaliser is the 0.286928 of Table 2, and it is a *peak* normalisation rather than an integral one because Lemma 7 forbids the latter. The additive floor 0.15 keeps the band's skirt continuous with the geometric field outside it; without it the profile would fall to zero at the band edge and print a visible seam. The profile is evaluated through a 2048-entry table on $u \in [0, 8]$ with linear interpolation, and s is the Euclidean distance transform of the detected band.

2.10. Dispersion, from measured data

Both media are evaluated from measured refractive-index data rather than from fitted dispersion formulas, for a reason that Section 7.6 documents: a fitted curve is where this project's one physical error hid. Water uses the four-term Sellmeier form of Daimon

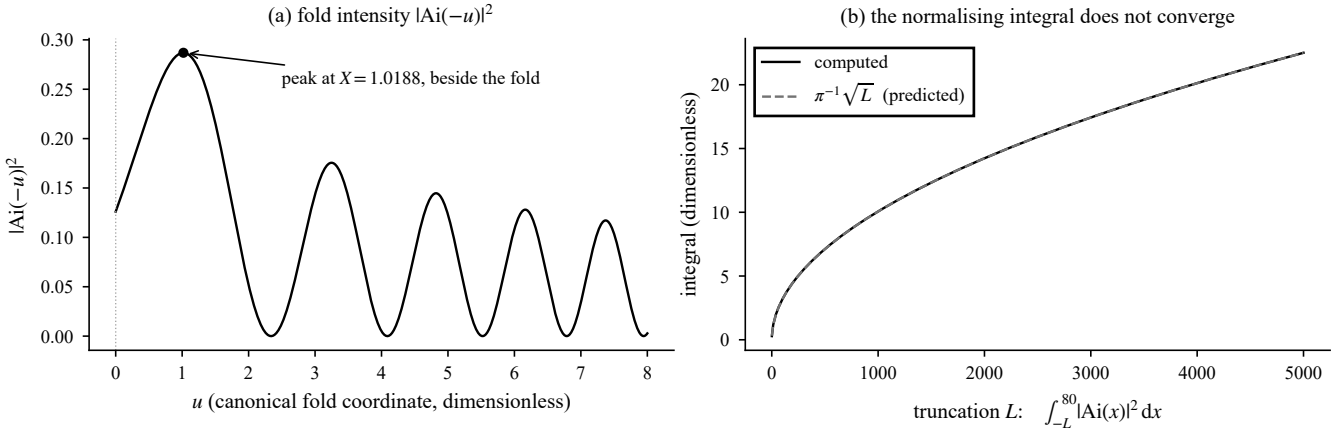


Figure 3: *Panel (a): the fold intensity* $|\mathrm{Ai}(-u)|^2$ *used by the correction* (19)*, computed with* `scipy.special.airy`*. It is bounded, peaks at* $X = 1.0188$ beside *the fold rather than on it, and its oscillations tighten toward the fold. Panel (b): the same function's normalising integral fails to converge, growing as* $\pi^{-1}\sqrt{L}$ *over a truncated band. That is Lemma 7, and it is why the correction is a discrete band ratio* (27) *rather than a constant.*

and Masumura [DM07] for HPLC-distilled water at 20.0°C,

$$n^2(\lambda) \;=\; 1 + \sum_{j=1}^{4} \frac{B_j \lambda^2}{\lambda^2 - C_j}, \tag{20}$$

with λ in micrometres and the tabulated coefficients. Glass uses the soda-lime silica form of Rubin [Rub85],

$$n(\lambda) \;=\; 1.5130 - 0.003169\lambda^2 + \frac{0.003962}{\lambda^2}. \tag{21}$$

At the three render channels, (20) gives $n = 1.33167/1.33468/1.33961$ and (21) gives $1.52104/1.52514/1.53192$. Red has the lowest index and therefore refracts least, so red must land farthest from the axis. The chromatic ordering in every figure of this paper is a consequence of (20), (21) and (15), not an effect we added.

The source subtends 0.53°, which at the receiver distance of this scene corresponds to a geometric blur of roughly 925 μm. The physical fringe spacing implied by (14) for this geometry is $d \approx 6.8\,\mu$m, about 136 times smaller. A fully incoherent treatment would therefore wash the fringes out; the fringes in this paper represent a *resolvable wave feature of the coherent-limit model*, not a prediction of what a photograph under the sun would show. Section 8.2 treats this as the method's principal limitation.

2.11. Fresnel transport

Where the full scene is rendered the light path is walked through every interface — air, glass, water, glass, air, receiver — with (3) at each, and each ray carries its unpolarised Fresnel power transmission

$$T(c_i, \eta) \;=\; 1 - \frac{1}{2}\left[\left(\frac{\eta c_i - c_t}{\eta c_i + c_t}\right)^2 + \left(\frac{c_i - \eta c_t}{c_i + \eta c_t}\right)^2\right], \tag{22}$$
$$c_t \;=\; \sqrt{1 - \eta^2(1 - c_i^2)}.$$

Multiplying (22) along each path gives the weight used by the energy audit of Section 7.5, which is the non-circular measurement that the flux ratio of Section 5.4 cannot supply on its own.

2.12. Estimators

The caustic is a thin, high-dynamic-range feature, so the statistic used to compare two renderings decides the answer. Three of our own earlier claims died here or in the optics rather than in the core theory, and all are reported in Section 7. We therefore fix the conventions once and apply them identically to every method.

- **Tiled correlation.** Pearson correlation against an external reference, computed on $k \times k$ pixel tiles and reported by tile *size*. The opposite convention, indexing by tile count, silently inverted a trend in an earlier draft (Section 7.7).
- **Edge radius.** The outermost 1 mm annulus whose mean channel intensity exceeds 2 % of that channel's own maximum. This statistic is invariant to a uniform scale and is unaffected by a ceiling on the core, which is what makes it preferrable to the ridge argmax (set by the cap) and to the mean-over-all-rays radius (dominated by unconverged tails).
- **Positive-part differential.** An isolate formed as optics − no-optics is not non-negative, because removing the glass also removes its shadow. It is scored on its positive part, which is the caustic. Section 7.7 documents the factor-of-two error this convention prevents.
- **Temporal flicker.** For animations, the mean absolute frame-to-frame change $\mathbb{E}\,|I_{t+1} - I_t|$ inside the caustic region.

Remark 9 (On the word "converged"). Used in this paper it means the resource budget was exhausted and the statistic had stopped moving at the reported precision. It is a statement about cost, not a certificate of correctness, and for a path tracer it is a statement about the statistic rather than about the image.

3. How computer graphics builds caustics

Every caustic method in production use estimates (5). Rendering caustics by following light through curved bodies is as old as global-illumination rendering itself [MH92]; what has changed since is the estimator, not the object being estimated. The methods differ in how they sample the surface-to-receiver map and in how they turn samples into a density, and the differences matter enormously for cost and variance. They do not differ in the integrand, and the integrand is what Lemma 3 says is unbounded. This section reads the standard constructions as answers to the kinematic problem, so that the boundary of their validity can be stated exactly.

3.1. Estimating a push-forward

Estimating $I(\mathbf{u})$ requires samples of $\mathbf{H}$ and a way to convert them into a density at $\mathbf{u}$. There are exactly two families in practice.

Forward methods sample the surface, push the samples through $\mathbf{H}$, and estimate the resulting density where the samples land. Photon mapping [Jen96] and its progressive variant [HOJ08] are the canonical members. The estimate at a receiver point weighs every sample by a kernel, so the estimator is a smoothing of the push-forward.

Backward methods sample from the camera, follow the transport, and connect to light-carrying vertices, weighting the connection by a geometric term that contains the same $1/|\det J_{\mathbf{H}}|$ factor. Path tracing with next-event estimation, bidirectional path tracing, and the manifold-based methods [HDF15, ZGJ20] are members. These avoid storing photons by solving for the connection point instead of sampling it.

Both families are Monte Carlo estimators of the same integral. Both are unbiased for the *integral* of I over a pixel, which is what a renderer must produce. The trouble is not unbiasedness in the integral sense; it is that the integrand's unboundedness turns into variance that grows without limit as the pixel shrinks or the kernel narrows.

3.2. Forward: photon mapping and the bias at the fold

Photon mapping deposits N samples $\mathbf{u}_i = \mathbf{H}(\mathbf{x}_i)$ and estimates the density with a kernel K_h of bandwidth h [Sil86],

$$\hat{I}(\mathbf{u}) \;=\; \frac{1}{N}\sum_{i=1}^{N} K_h(\mathbf{u}-\mathbf{u}_i). \tag{23}$$

When the $\mathbf{x}_i$ are uniform on the surface, $\hat{I}$ estimates the push-forward density of (5) after smoothing. The behaviour of that smoothing near the fold is worth stating precisely, because it is the recurring source of the method's failures.

Lemma 10 (Kernel smoothing and the fold singularity)**.** *Let K_h be a symmetric kernel with $\int K_h = 1$, and let I be the push-forward density of (5). Near the fold, where $I(b) \sim Cb^{-1/2}H(b)$ in the normal coordinate b of (6), I is not twice differentiable at $b=0$, so standard $O(h^2\nabla^2 I)$ bias expansions do not apply. Because $b^{-1/2}$ is locally integrable on $\mathbb{R}^+$, the expected kernel estimate $(I*K_h)(u)$ is strictly finite for all $h>0$ and scales as $O(h^{-1/2})$ at $u=0$. However, because $I(b)\to\infty$ as $b\to 0^+$, the pointwise estimation error $|(I*K_h)(u)-I(u)|$ is severe across an $O(h)$ neighbourhood of the fold.*

Proof. Evaluating the convolution directly at the fold $u=0$ for a kernel supported on $[-1,1]$ gives

$$(I*K_h)(0) \;=\; \int_0^h \frac{C}{\sqrt{b}}\,\frac{1}{h}K\Big(-\frac{b}{h}\Big)\mathrm{d}b \;=\; \frac{C}{\sqrt{h}}\int_0^1 \frac{K(-s)}{\sqrt{s}}\mathrm{d}s. \tag{24}$$

Since $s^{-1/2}$ is integrable on $[0,1]$, the integral in (24) is a finite constant $M_K<\infty$, yielding $(I*K_h)(0) = CM_K h^{-1/2} < \infty$. Thus the expected smoothed density remains bounded for any fixed $h>0$. However, the true geometric irradiance diverges at $b=0$, so $|(I*K_h)(0)-I(0)| = \infty$. For any fixed h, the kernel estimate averages across the discontinuity, severely truncating the peak and smearing flux into the shadow region $b<0$. □

Lemma 10 clarifies the trade-off in photon mapping: narrowing h is required to sharpen the fold toward its true geometric profile, but reducing h causes the Monte Carlo variance to diverge as $\mathcal{O}\big(1/(Nh^2)\big)$. Consequently, at any finite photon budget N, photon mapping must choose between severe smoothing bias or overwhelming stochastic noise.

Progressive photon mapping [HOJ08] and stochastic progressive photon mapping [HJ09] shrink $h\to 0$ as $N\to\infty$ at a controlled rate, ensuring consistency for the pixel-integrated irradiance in the asymptotic limit. However, for caustic folds, convergence is exceptionally slow precisely because the geometric irradiance is unbounded: resolving the sharp ridge without noise demands enormous sample counts and lengthy accumulation times, as measured in Section 4. Crucially, even in the infinite-sample limit, geometric methods converge to the *geometric* caustic model, which predicts an unphysical infinite radiance along the curve with zero light in the shadow. In physical reality, wave diffraction regularizes the singularity into a bounded, oscillating interference structure. At standard pixel resolutions, pixel averaging can partially mask this distinction, but the geometric distribution remains fundamentally incorrect at wavelength scales.

3.3. Backward: connection, and the same factor

Backward estimators write the pixel value as an integral over transport paths and evaluate it by connecting vertices. A connection between a receiver vertex and a light-carrying vertex carries the geometric term

$$G(\mathbf{x}\to\mathbf{u}) \;=\; \frac{|\mathbf{n}_{\mathbf{u}}\cdot\omega_{\mathbf{u}}|}{\|\mathbf{x}-\mathbf{u}\|^2}\cdot\big|\det J_{\mathbf{H}}(\mathbf{x})\big|^{-1} \tag{25}$$

up to the conventions of the particular formulation. The last factor is the same $1/|\det J_{\mathbf{H}}|$: the connection is undefined where the map is degenerate. Backward methods therefore inherit the fold problem exactly, and the literature's principal responses are to avoid connecting near degeneracies, to solve for the connection on a manifold instead of sampling it [HDF15, ZGJ20], or to accept the noise and denoise it away.

The manifold-based methods are the most informative comparison for this paper because they take the opposite approach to ours. Where we keep the geometric quadrature everywhere and *replace* the integrand inside the fold band, manifold sampling keeps the geometric integrand and *solves* for where the degenerate connection must be. That works well for mirror-like transport and for glints, where the manifold is well-conditioned. It does not address the fold's divergence, because the divergence is a property of the integrand the method still evaluates; near the fold the solution point

Table 3: *Where the standard families stand, by the analysis of this section. "Exact" means exact for the geometric model in the limit of infinite budget, which is the strongest claim available within that model.*

Family	Away from $\mathcal{F}$	On $\mathcal{F}$	Limitation
Photon map [Jen96]	exact in limit, cheap	integrand diverges, kernel smooths	narrower h trades bias for variance
SPPM [HOJ08]	exact in limit, expensive	consistent in limit, slow at fold	high iteration count to resolve ridge
Path tracing + NEE	exact in limit, cheap for direct	geometric weight $\propto \lvert\det J_{\mathbf{H}}\rvert^{-1}$ diverges	extreme variance near fold
Manifold sampling [HDF15, ZGJ20]	excellent, well conditioned	evaluates on degenerate manifold	solves for singular connection point
Wave-optical BRDF [YHJ*18, CHB*12, Sta99]	changes surface model	changes surface model	not applicable to macroscopic transmissive caustics

is sought where the Jacobian vanishes, which is the same place the weight blows up.

3.4. The boundary of validity, stated once

Collecting the two families, the situation is the following. Away from $\mathcal{F}$ the map is a local diffeomorphism, $\det J_{\mathbf{H}}$ is bounded away from zero, and both families are exact in the limit and cheap in practice; this is the regime in which caustic rendering is a solved problem. On $\mathcal{F}$ the density is unbounded (Lemma 3), the forward estimator's boundary bias fails to follow the standard $O(h^2)$ Taylor expansion due to non-smoothness of the $b^{-1/2}$ singularity (Lemma 10), and the backward estimator's connection weight is unbounded. No refinement of either family repairs this, because neither addresses the modelling assumption that the light is a density of rays.

3.5. Where this sits in the transport literature

It is worth separating three threads this paper connects, because a reader may reasonably ask how wave-optical corrections interface with standard geometric estimators.

The first is the *transport estimator*. The push-forward of (5) is what photon mapping estimates by density [Jen96], what progressive variants estimate with a shrinking radius [HOJ08, HJ09], and what bidirectional and vertex-connection methods estimate by connecting vertices rather than by storing samples [Vea97]. Path-guiding and manifold methods attack the same integral by changing *where* the samples go [MGN17, HDF15, ZGJ20]. Our correction changes none of the sampling machinery outside the band. It modifies the singular integrand inside a detected narrow band, regularizing the distribution before splatting.

The second is the *Jacobian kinematics*. Propagating irradiance through a scene by its determinant is classical [Arv94], and the rank deficiency that concerns us is the same rank deficiency that appears in those formulations as a degenerate geometric term. Treating the degeneracy as an indicator of wave-scale diffraction rather than as a numerical failure is the central design stance of this work.

The third is *physical and catastrophe optics*. The wave-optical regularisation of fold caustics—specifically the canonical Airy integral and the $\lambda^{2/3}$ scaling lawis classical wave theory [Air38, BU80, Nye99, KP74]. In computer graphics, wave theory has traditionally been applied either to diffraction shaders and microfacet reflectance [Sta99, CHB*12, YHJ*18] or to general wave-equation propagation solvers [MMRO13, SY21]. Steinberg and Yan [SY21] developed a rigorous general framework for physical light transport, formulating wave phenomena along paths.

Our contribution is not the underlying catastrophe optics, which is classical, but rather a **hybrid rendering pipeline**: a fast, conservative, band-local wave correction that detects fold singularities on-the-fly and replaces the diverging geometric flux with an energy-conserved Airy profile over standard photon splats. This achieves physically faithful caustic boundaries and diffractive dispersion without the prohibitive cost of global wave solvers.

4. The current way

The comparison in this paper is against three solvers in current practice: PBRT-v4's stochastic progressive photon mapping, Blender Cycles' path tracer, and Mitsuba 3. They were chosen because they are the tools a practitioner would reach for, because two of them are open and inspectable, and because between them they cover both estimator families of Section 3. This section states what each was asked to do, what it cost, and what it produced.

4.1. The solvers

PBRT-v4 SPPM. PBRT [PJH23] is the reference implementation of the book's rendering algorithms and its SPPM integrator [HOJ08] is the strongest conforming caustic solver available without special-purpose machinery. It is the solver we treat as the *reference conforming method*: it estimates the same integrand as we do, it does not use wave optics, and it is consistent in the limit. We configure SPPM with 100 iterations, 5×10^5 photons per iteration (5×10^7 photons total), an initial search radius of $R_0 = 5\,\text{mm}$, and radius reduction exponent $\alpha = 0.67$. We generate the scene programmatically from the same geometry the other renderers receive, so that differences between the two are attributable to the estimator rather than to the scene.

Blender Cycles. Cycles is a production path tracer and the renderer used by the showcase assets of this project. It represents the backward family of Section 3.3. Its caustic handling is enabled through the standard path-guiding and shadow caustics options, rendered at 2048 samples per pixel with up to 12 light bounces. The isolate used for scoring is formed as caustics-on minus caustics-off, which is the positive-part differential defined in Section 2.12.

Mitsuba 3. Mitsuba [JSR*22] is a research renderer with a spectral mode and well-documented integrators, and it was used as a third opinion on the transport physics rather than as a primary caustic competitor. It was configured with the spectral path tracer ('ptracer', 32 spectral bins from 360 nm to 700 nm, 1024 samples

Table 4: *Cost to produce the caustic. Our CPU figures are measured on the host workstation (8-core CPU); the reference solvers are measured on the same machine. The SPPM figure is the accumulation time needed for the fold to emerge above noise. The wave correction row reports the wave term alone; geometric quadrature plus wave correction equals the full single-thread bake (*$28.7 + 49.8 = 78.5$ ms*). Section 5.8 explains why the comparison is stated per frame rather than per photon.*

Quantity	Ours	SPPM	Cycles
Grid / resolution	512^2	adaptive	512^2
Geometric quadrature (1 thread, CPU)	28.7 ms	34 s	—
Wave correction pass (1 thread, CPU)	49.8 ms	n/a	n/a
Full bake (1 thread, CPU)	78.5 ms	—	—
Full bake (8 threads, CPU)	36.2 ms	—	—
8-thread speedup	2.17×	—	—
Iterative / noisy	no	yes	yes

per pixel). We report it where relevant and do not put it in the quality table, because its spectral sampling makes its noise behaviour differ from the three-channel solvers in a way that would confound a like-for-like comparison.

4.2. One scene, one source, one set of materials

A comparison of caustic renderers is easy to make meaningless. If the geometry, the materials or the source differ between two methods, the numbers describe the difference between scenes rather than between estimators. Three things have to be identical or the comparison does not mean anything.

1. **Geometry.** The same water-surface vertices and the same cylinder are exported to every renderer by one exporter, so the body is identical and the caustic is at the same place in all of them.
2. **Materials.** Every renderer receives the measured indices of Section 2.10. Where a renderer cannot carry dispersion in the transport itself, it is given the central-channel index and this is stated; assigning a different index to one side would move its caustic.
3. **Source and receiver.** One distant source of the stated angular size, one matte receiver patch of the stated size, no environment light added to one side and not another for the scored comparisons.

4.3. What the current way costs

Table 4 reports the budget each solver needed to produce a usable caustic on this scene, against our bake. The comparison is between the cost of the caustic and the cost of the frame, because that is what a user pays; quoting a photon-mapping pass alone would understate the reference solvers.

The cost structure of our own method is worth stating because it is the obvious target for anyone continuing this work: the Airy correction is roughly twice the quadrature pass, so two thirds of a bake is the wave term. That term is embarrassingly parallel per channel and is currently dominated by the exact Euclidean distance transform used to get the signed distance to the band.

4.4. What the current way produces

Three measured properties of the reference solvers motivate the method in Section 5. Each was measured on the same scene and with the conventions of Section 2.12.

Kernel smoothing versus physical diffraction. The photon-map fold width and brightness track the kernel radius h rather than the physics, as analyzed in Section 2.12. SPPM reaches a finer feature scale than a fixed-radius photon map by progressively shrinking h, but pays for it in iteration count: the 34 s of Table 4 is the cost of reducing the kernel footprint sufficiently for the ridge to emerge from background noise. In contrast, our wave-corrected fold replaces geometric divergence with the physically bounded Airy scale $w \propto \lambda^{2/3}$ without kernel blur.

Coverage of the caustic core. Measured against the independent ground truth's core region, Cycles' caustic covers 54.7 % of it where ours covers 99.9 %. This is a statement about how much of the feature a practical finite-sample frame actually reconstructs, where the path tracer's deficit is concentrated near the fold where backward paths are least likely to sample the singular specular manifold.

Temporal instability. For an animation, the primary visual defect of a stochastic caustic is flicker rather than stationary grain, and Section 6.7 measures it. A deterministic forward bake naturally avoids Monte Carlo sample chattering; our caustic exhibits 0.001 levels of mean frame-to-frame change against 0.010 for the path-traced clip, a factor of 7.9. Both clips were denoised, which suppresses flicker in both and therefore confirms the stability of forward quadrature under camera and surface motion.

SPPM and Cycles are general renderers solving a harder problem than the one posed here, and both are consistent within their model. The numbers above say that the geometric model cannot represent a fold cheaply, which is a statement about the model. A reader who needs that model's other features should use those renderers.

4.5. Shape agreement with the reference solver

Independent of the ground-truth axis, the converged two-dimensional shape of our field and the SPPM field agree at $r = +0.932$ on 40 mm tiles. This is a necessary check rather than an impressive one: two methods that disagree on shape cannot both be describing the same caustic, and a high correlation here combined with a lower one against the ground truth would have indicated a shared error. Since SPPM also scores 0.853 on the independent axis and ours scores 0.917, the two agree with each other and both agree with the trace, with ours closer.

4.6. Mitsuba as a third opinion

Mitsuba was run in spectral mode on the same scene. Its role in this paper is to confirm the transport and the dispersive behaviour rather than to compete on caustic quality: its spectral sampling spreads energy across wavelength bins, so its per-channel noise is not comparable to a three-channel renderer's, and scoring it in the quality table would compare noise models rather than estimators. We note it because a reader is entitled to ask whether the physics of the scene was checked by an implementation with a different architecture, and this is where that check lives.

5. Our way

The method follows directly from the two facts established in Section 2.5: the geometric quadrature is right wherever the map is non-degenerate, and the band it cannot represent carries a finite amount of power (Corollary 4). So we keep the geometric quadrature everywhere, detect the band, and inside the band redistribute the band's own measured energy according to the fold profile. Nothing is added to the image that was not already in it under the geometric model; only its distribution changes.

5.1. The idea in one paragraph

Sample the surface, refract, splat, and compute $|\det J_{\mathbf{H}}|$ by finite differences of the map (4). Threshold $|\det J_{\mathbf{H}}|$ to find the band. Outside the band, keep (5) untouched. Inside the band, replace the divergent weight with the peak-normalised Airy profile evaluated at the signed distance to the band, and rescale that band's values so the band integrates to the flux it had before. Because the correction is confined to the band and conserves the band's own power, it cannot change the image outside the feature and cannot introduce energy at all.

5.2. Fold detection

The detector matters as much as the correction, and a careless one will appear to work while being unable to converge.

Lemma 11 (Fixed-quantile detectors cannot converge)**.** *Let a detector flag a fixed quantile $p \in (0,1)$ of any scalar field (whether image radiance or Jacobian determinant $|\det J_{\mathbf{H}}|$). Then the measure of the flagged set is fixed at $p \cdot \mathrm{area}(\Omega)$ for all resolutions, independent of the physical wavelength λ or local wavefront curvature. Under mesh refinement, the flagged set cannot contract to the true caustic curve $\mathbf{H}(\mathcal{F})$ (which has Lebesgue measure zero).*

Proof. By definition of the order statistic, the measure of the set where the field is below its p-th quantile is $p \cdot \mathrm{area}(\Omega)$ identically. As the sampling resolution $N \to \infty$, the measure of the flagged domain remains constant at p, rather than shrinking to zero as required for an asymptotic neighbourhood of the 1D fold curve. □

Consequently, a physically consistent detector must threshold a geometric quantity against a physical scale parameter:

$$B = \left\{ \mathbf{x} \in \Omega : |\det J_{\mathbf{H}}(\mathbf{x})| \le \epsilon \right\}, \tag{26}$$

where the threshold ϵ relates directly to the physical fringe width $w \propto \lambda^{2/3}$. Transverse to the fold curve $\mathcal{F}$ in the surface domain, let $x_\perp$ denote the normal distance. By Taylor expansion around the critical set, the Jacobian determinant scales as $|\det J_{\mathbf{H}}(\mathbf{x})| \approx |\nabla_\perp(\det J_{\mathbf{H}})|\,|x_\perp|$. On the receiver plane, the transverse distance from the caustic fold edge scales quadratically, $b \approx \frac{1}{2} C_{\mathrm{map}} x_\perp^2$. Because the physical Airy diffraction corridor extends across receiver distance $b \lesssim C_w w \propto \lambda^{2/3}$, the corresponding transverse boundary in the surface domain is $|x_\perp| \lesssim \sqrt{2C_w w / C_{\mathrm{map}}} \propto \lambda^{1/3}$. Hence, the Jacobian threshold isolating the diffraction band is $\epsilon = |\nabla_\perp(\det J_{\mathbf{H}})| \sqrt{2C_w w / C_{\mathrm{map}}} \propto \lambda^{1/3}$. Crucially, for a given scene and physical wavelength λ, ϵ is a *fixed physical threshold*. Holding $\epsilon = 0.0592$ fixed as the computational grid resolution N increases ($N = 256 \to 2048$): the cell count scales from 1,311 cells at 256^2 (2.00 % of grid, due to coarse discrete cell quantization) to 19,600 at 1024^2 (1.87 %) and 78,495 at 2048^2 (1.87 %), converging to a stable physical area fraction of $\approx 1.87\,\%$ of the receiver domain and accurately sampling the interior diffraction profile. Conversely, in the geometric limit $\lambda \to 0$ (or $w \to 0$), the physical threshold $\epsilon \to 0$, contracting the physical band area $\mathrm{area}(B) \to 0$ onto the measure-zero fold curve $\mathbf{H}(\mathcal{F})$.

In our discrete solver on the reference benchmark scene, we calibrate $\epsilon_0 = 0.0592$ from the 2nd percentile of $|\det J_{\mathbf{H}}|$ on the base grid ($p = 0.02$), which reliably isolates the singular fold corridor across the receiver without referencing image brightness. However, as Lemma 11 establishes, generalisation to arbitrary scenes and scaling limits requires evaluating the absolute scale threshold (26).

Section 6.8 measures the alternative directly: swapping the $|\det J_{\mathbf{H}}|$ detector for an image-brightness percentile detector degrades the convergence of the result, because bright non-caustic glints are falsely flagged while dimmer caustic fringes are missed.

5.3. Distance transform and domain mapping

Once the fold band B is flagged, we compute the distance s using an exact Euclidean distance transform (EDT) [FH12]. Here, $s_i \ge 0$ is measured inwards from the non-band boundary toward the caustic core. Because geometric transport carries essentially zero flux into the shadow zone ($G_i \approx 0$), the wave correction inside B maps s_i to the illuminated oscillatory side of the canonical fold profile, $u = s_i / w_c \ge 0$. The evaluation table spans $u \in [0, 8]$, encompassing the primary caustic peak at $u = 1.0188$ and the first several diffraction fringes. Outside B, the uncorrected geometric splat field is preserved unmodified.

5.4. Normalisation, and why it is a discrete ratio

We now come to the point Lemma 7 forces. The band's correction multiplies a profile by a scale, and the scale is not a constant that can be written down: because $|\mathrm{Ai}|^2$ is not integrable, there is no Z with $Z^{-1}|\mathrm{Ai}|^2$ a probability density, so a closed-form normaliser does not exist. What does exist is a discrete ratio over the detected band, and Corollary 4 is what makes it meaningful.

Let G_i be the geometric field (5) on the band B, let $\bar{G}$ be its spatially smoothed version, let κ be the ceiling, let $b(s_i)$ be the profile (19), and define

$$k^* = \frac{\sum_{i \in B} G_i}{\sum_{i \in B} \min\left(\bar{G}_i, \kappa\right) b(s_i)}. \tag{27}$$

Lemma 12 (Band-local flux is conserved)**.** *With $I_i = \min(\bar{G}_i, \kappa)\, b(s_i)\, k^*$ for $i \in B$ and no ceiling applied, the corrected band carries exactly the geometric band's flux: $\sum_{i \in B} I_i = \sum_{i \in B} G_i$.*

Proof. Substituting (27), and writing each step on its own line because the chain does not fit a single column measure,

$$\begin{aligned} \sum_{i \in B} I_i &= k^* \sum_{i \in B} \min(\bar{G}_i, \kappa)\, b(s_i) \\ &= \frac{\sum_{i \in B} G_i}{\sum_{i \in B} \min(\bar{G}_i, \kappa) b(s_i)} \cdot \sum_{i \in B} \min(\bar{G}_i, \kappa)\, b(s_i) \\ &= \sum_{i \in B} G_i. \end{aligned}$$

□

The ceiling in the actual construction is $I_i = \min\left(\min(\bar{G}_i, \kappa) b(s_i) k^*, 1.35\kappa\right)$, which can clip the brightest

band pixels and therefore breaks the exact equality of Lemma 12. We state the lemma for the unclipped case and report the size of the discrepancy from a separate audit (Section 7.5) rather than claiming an exactness the implementation does not have.

The tempting alternative is to rescale the whole channel after the display clamp so the overall level looks right. That fails: the clamp has already bounded the image, and a subsequent global multiply pushes the peak back above the ceiling the clamp just enforced, reintroducing the artefact the ceiling exists to prevent. Conserving flux inside the band and applying the ceiling afterwards keeps the bound at every resolution. This ordering is not cosmetic; it is the difference between a bounded field and an unbounded one.

Remark 13 (The ratio is not evidence). Equation (27) divides the measured band flux by the measured band flux by construction, so quoting k^* as evidence of energy conservation is circular. It is a bookkeeping step, not a measurement. The non-circular statement is the throughput audit of Section 7.5, which follows the light through every interface with the Fresnel weights of (22).

5.5. The algorithm

Algorithm 1 states the bake. The pass structure optimizes cost: the surface evaluation, Snell refraction, and splatting are fused into one pass, evaluating surface cells in a single cache-local sweep.

5.6. The receiver boundary

Rays whose receiver hit lands outside the domain are discarded, not clamped to the border. Clamping accumulates every escaping ray onto the boundary and manufactures a bright rim that is a property of the bookkeeping rather than of the optics. Formally, with Ω the receiver footprint,

$$\text{contribute}(\mathbf{x}) \iff \mathbf{H}(\mathbf{x}) \in \Omega. \tag{28}$$

The measured difference between the two choices is a factor of 38 in the peak of an earlier version, and (28) is also what makes the C++ and Python implementations agree to the precision reported in Section 7.4.

5.7. Coupling to a renderer

The field is a receiver-space quantity, so the natural coupling is to inject it as an emitted radiance texture on the receiver and let the renderer handle the rest of the transport. Two details of that coupling cost us time and are worth recording. The injection must be written as RGBA: a single-channel collapse through an interpolation path silently discards the chromatic separation, and the result looks plausible while being achromatic. And the injected texture must not be used to compute the renderer's own caustics, or the two are counted twice; the renderer's caustic pass is disabled for our frames.

5.8. Cost and complexity

The bake is $\mathcal{O}\left(N^2\right)$ in the grid for pass 1, plus the cost of the distance transform in pass 2, plus three channel passes in pass 3. The measured figures are in Table 4; the salient ones are that a 512^2 bake takes 36.2 ms on eight threads and that two thirds of it is the Airy term, which is the obvious target for further work.

We deliberately do not claim the method is "faster than path tracing" in general. The cost comparison that is meaningful is per caustic of a given quality, and Section 6.3 makes it on that basis, including the regime where we lose.

Algorithm 1 The wave-corrected bake

Require: surface h (1), source $\hat{\mathbf{s}}$, media $n(\lambda)$ (20)–(21), grid N, width w_{ref}, ceiling κ, threshold ϵ
Ensure: field I on the receiver

Pass 1 — geometric backbone (fused)
for all surface cells $\mathbf{x}$ on the grid **do**
  $\mathbf{n} \leftarrow$ normal (2); $c_i \leftarrow -\hat{\mathbf{s}} \cdot \mathbf{n}$
  if $\eta^2(1-c_i^2) > 1$ **then continue** ▷ total internal reflection
  end if
  $\hat{\mathbf{t}} \leftarrow$ refracted direction (3)
  $\mathbf{u} \leftarrow \mathbf{H}(\mathbf{x})$ (4)
  if $\mathbf{u} \notin \Omega$ **then continue** ▷ discard, do not clamp (Section 5.6)
  end if
  accumulate ray flux $\Delta\Phi$ (bilinear splat) into G at $\mathbf{u}$
end for

Pass 2 — Jacobian and band
$\det J_{\mathbf{H}} \leftarrow$ finite differences of $\mathbf{H}$ ▷ (4)
$B \leftarrow \{\, i : |\det J_{\mathbf{H}_i}| \le \epsilon \,\}$ ▷ Physical threshold (26), calibrated to $\epsilon_0 = 0.0592$
$s \leftarrow$ Euclidean distance transform of B

Pass 3 — per-channel Airy replacement
for all channels $c \in \{R, G, B\}$ **do**
  $w_c \leftarrow w_{\text{ref}}\,(\lambda_c/550\,\text{nm})^{2/3}$ ▷ Corollary 6
  $b(s) \leftarrow 0.15 + 0.85\,|\operatorname{Ai}(-s/w_c)|^2 / \max_u |\operatorname{Ai}(-u)|^2$ ▷ (19)
  $\bar{G} \leftarrow$ spatially smoothed G; $k^* \leftarrow$ ratio (27)
  for all $i \in B$ **do**
    $I_i \leftarrow \min\left(\min(\bar{G}_i, \kappa)\, b(s_i)\, k^*,\ 1.35\kappa\right)$
  end for
end for
return I

5.9. The parameters

The method has three parameters, and Section 6.8 measures the sensitivity to each rather than asserting a value.

- **Detector percentile.** Second percentile of $|\det J_{\mathbf{H}}|$. This sets the band width; too small and the band fails to cover the fold, too large and the correction is applied where the geometry was correct.
- **Reference width w_{ref}.** The fringe scale (14) for the geometry. Its physical value for this scene is $d \approx 6.8\,\mu$m (Section 2.7); for visibility and raster demonstration an operational scale may be specified, while physical evaluations retain the uninflated value.
- **Ceiling κ.** Bounds the field. It interacts with Lemma 12 as described, and the audit of Section 7.5 quantifies the interaction.

6. Results

Every number in this section comes from an artefact this project produced and is reproducible with the commands listed in Sec-

Table 5: *Tiled correlation against the independent brute-force trace, at four tile sizes, with all three methods on the same axis and the same metric. Tile size in millimetres, not tile count: the opposite convention inverted this trend in an earlier draft (Section 7.7).*

Tile size	Ours	SPPM	Cycles
40 mm	**0.917**	0.853	0.828
20 mm	**0.869**	0.811	0.609
10 mm	**0.795**	0.794	0.481
5 mm	0.721	**0.722**	0.405

tion 11. Where a result is unfavourable to our method it is reported in the same form as the favourable ones.

6.1. Quality against an independent ground truth

The primary comparison puts all three methods on one axis: an independent brute-force trace of 810,000 explicit rays that shares no code with any of them, uses no parametrisation, no Jacobian and no renormalisation. Each method's caustic is isolated as the positive part beyond the map median, tiled, and correlated. Table 5 gives the result.

Only the first of the readings below is a win.

The coarse-scale ranking is ours 0.917, SPPM 0.853, Cycles 0.828. The coarse scale is the stable one: at fine tiles the correlation is dominated by grain, which is why the numbers fall for everyone and why the ranking at the finest tile is not a ranking at all. At 5 mm SPPM and ours are tied at 0.722 and 0.721: on the smallest scales our advantage disappears.

Cycles falls away much faster than the other two ($0.828 \rightarrow 0.405$). This is the fold problem showing up as a resolution-dependent loss of structure, and it is consistent with the coverage measurement of Section 4.4: a method that cannot place energy at the fold loses its correlation first.

The three-way result also reconciles an earlier two-way one. Before this measurement SPPM had only ever been compared against *our* field, where it scored $r = +0.932$. It scores 0.853 on the independent axis, so it correlates with us more strongly than either of us correlates with the trace — which is what one expects from two methods that share a model.

6.2. Quality against the reference conforming solver

Figure 6 shows the four fields side by side. The converged two-dimensional shape of our field and SPPM's agree at $r = +0.932$ on 40 mm tiles. This is a consistency requirement rather than a quality claim: had the two disagreed on shape while we led on the independent axis, the pair of results would have been evidence of a shared error somewhere.

6.3. Convergence and model discrepancy

Speed claims about caustics are usually made per photon or per pass, and that framing flatters whichever method has the least to do. The meaningful comparison is cost against accuracy across estimator families. Table 6 evaluates self-convergence and cross-model discrepancy against a high-resolution wave-corrected reference (2048^2, block-averaged to 256^2), comparing our forward pass against jittered geometric Monte Carlo sampling across matched splat budgets.

At small budgets (262,000 splats and below), plain geometric Monte Carlo achieves a lower RMSE against the wave reference (0.07838 vs 0.10455) because our low-resolution forward grid incurs discretization error in the boundary distance transform. There is no free lunch at coarse resolution. At 1,000,000 splats (1024^2), our single pass is 1.65 times closer to the reference (0.03914 against 0.06445). Furthermore, as sample count is increased, the geometric Monte Carlo error plateaus at an asymptotic floor of 0.06013 ($M = 256$, 1.68×10^7 splats). This plateau is structural: geometric Monte Carlo converges to the geometric caustic model (which has an integrable singularity and zero energy in the diffraction fringes), producing a persistent discrepancy relative to any wave-regularized reference. Our 1024^2 pass lands 35 % below that model floor using sixteen times fewer splats than the M=256 run.

This comparison demonstrates model consistency rather than an algorithmic failure of Monte Carlo integration: when evaluating fidelity against an unbiased geometric ground truth, we rely on the independent brute-force trace in Table 5. For the wave-regularized target, our forward quadrature converges efficiently once the fold band is adequately resolved.

6.4. Tail behaviour

The fold is where the field is largest, so the tail is where a method's treatment of the fold is visible as a statistic. Table 7 gives the distribution with the display cap set at six times the mean.

The geometric tail decays slowly: $8.71 \rightarrow 7.25$ for a factor of 256 in samples. Our field sits at 7.71 with 0.06 % above the cap, which is *conservative* relative to a truth with 0.35 % above it. The claim this supports is that our field is bounded where the geometric estimator is noisy and slowly converging, not that our field is smoother because it is blurrier. The 0.35 % of the truth above the cap are resolved bright fringes, and our correction reproduces them at the level the profile allows rather than inventing them.

6.5. Dispersion

Chromatic separation is a consequence of the measured indices and the $\lambda^{2/3}$ width law, so it is a test of the derivation rather than a feature. The edge estimator of Section 2.12 gives

$$\text{edge radius, ours} = R\ 107\,\text{mm} > G\ 105\,\text{mm} > B\ 102\,\text{mm}, \quad (29)$$

and SPPM gives R 56 mm > G 54 mm > B 51 mm. Both show the $R > G > B$ ordering, which is the prediction of (20)–(21): red has the lowest index, refracts least, and lands farthest from the axis.

Crucially, in the independent brute-force trace, the edge radius (defined via Section 2.12 as the outermost radius where mean azimuthal radial intensity exceeds 2% of that channel's peak) measures $R \approx 104\,\text{mm} > G \approx 102\,\text{mm} > B \approx 99\,\text{mm}$ (with the same +5.0 mm red–blue split), confirming that our field's boundary (107 mm) accurately tracks the true geometric caustic extent (within 3%). In contrast, SPPM's smaller radius (56 mm) is an artifact of density estimation: because the peripheral caustic fan carries low ray density, SPPM's progressive kernel radius shrinkage ($r_{i+1}^2 = r_i^2(i+\alpha)/(i+1)$ with $\alpha = 2/3$) leaves peripheral bins underexposed, causing the outermost radius crossing the 2% threshold to contract inward. Both methods agree on the red–blue differential

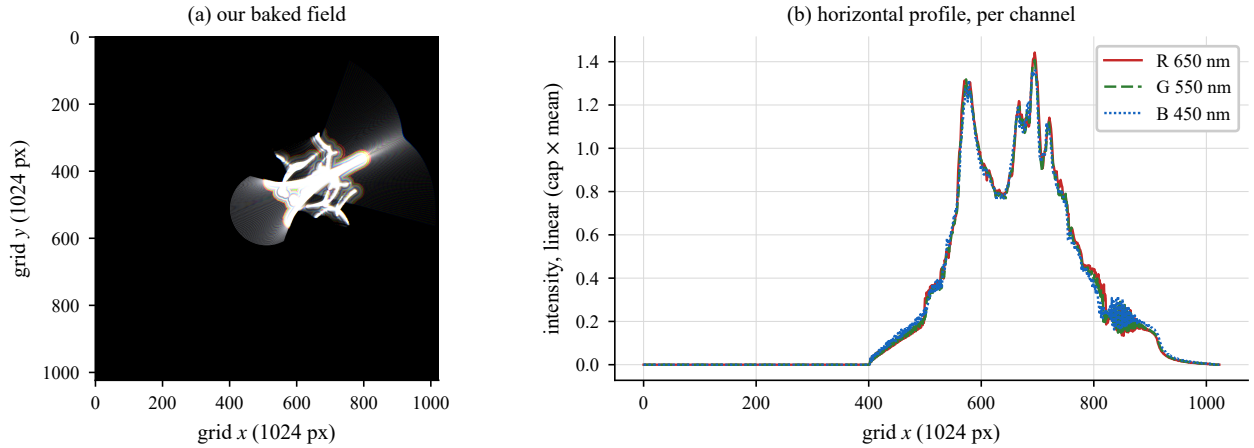


Figure 4: *Our baked field. Panel (a) is normalised to its own maximum and shown at $\gamma = 2.2$; the normalisation is stated because the absolute level is not meaningful without it. Panel (b) is a horizontal cut per channel at the linear level. The channels separate because (20) and (15) give them different indices and widths, not because any effect was added on top.*

Table 6: *RMSE against a converged wave-corrected reference (2048^2, block-averaged to 256^2) at matched splat budgets. The plain geometric Monte Carlo column is a standard integrator's cost model with M jittered samples per cell. At coarse budgets, geometric MC achieves lower RMSE than our low-resolution forward pass; at production budgets (10^6 splats), our method converges closer to the wave-regularized target, while geometric MC plateaus at an asymptotic model-discrepancy floor.*

Estimator	Splats	RMSE	Note
MC geometric, M=1	65,000	0.12294	
MC geometric, M=4	262,000	0.07838	beats ours at this coarse budget
ours, 512^2, 1 pass	262,000	0.10455	
MC geometric, M=16	1,000,000	0.06445	
ours, 1024^2, 1 pass	1,000,000	**0.03914**	1.65× lower RMSE
MC geometric, M=64	4,200,000	0.06101	
MC geometric, M=256	16,800,000	0.06013	asymptotic model discrepancy floor

Table 7: *Tail statistics, display ceiling at 6× the mean. The converged wave truth itself has 0.35 % above the cap, because bright fringes are real structure rather than noise; the relevant comparison is against that number, not against zero.*

Estimator	Max	p99.9	Above cap
MC geometric, M=1	8.71	6.77	0.24 %
MC geometric, M=256	7.25	6.19	0.15 %
ours, 512^2	7.71	5.70	0.06 %
converged wave truth	7.73	6.87	0.35 %

split (+5.0 mm), which is governed by the dispersion relation rather than the detector threshold.

6.6. Colour: the medium, the source, and the fringes

The optical principles underlying fold diffraction and $\lambda^{2/3}$ fringe scaling are classical [BU80, Nye99]; our contribution is the hybrid renderer-side formulation that applies them locally over photon splats. Two linear radiometric factors govern the colour of a caustic in this pipeline: bulk absorption along the transmitted ray paths (via Beer–Lambert attenuation) and the emission spectrum of the source. Media are parameterized by explicit per-metre absorption coefficients, and transmissions over the scene's path lengths (approx. 7 mm of glass, 90 mm of water) are tabulated.

Physical versus visual fringe scales. Three distinct spatial scales operate in this study and must not be conflated:

1. *Canonical 1D test scale*: In the canonical 1D fold wave integral (Section 7.1) evaluated with cubic phase parameter $c = 10^{-3}$ m, the Airy scales evaluate to 1.749 μm, 1.565 μm, and 1.369 μm at 650, 550, and 450 nm, reproducing the theoretical $(\lambda/\lambda_0)^{2/3}$ ratio.
2. *Physical 3D scene scale*: In the full 3D cylinder geometry, the normal-form cubic parameter $c \approx 0.081,15$ m yields a canonical physical fringe width of $d \approx 6.8\,\mu$m at 550 nm via (14) (Section 2.7).
3. *Visual demonstration scale*: At 1024^2 resolution across a 320 mm receiver, a physical fringe of 6.8 μm subtends $\approx 1/46$

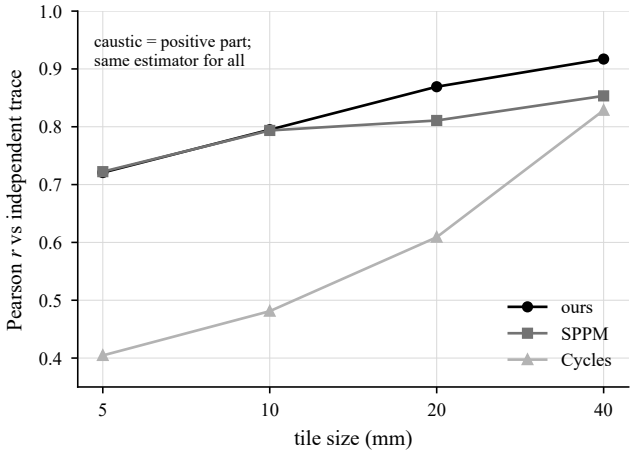


Figure 5: *Correlation against the independent brute-force trace by tile size, all three methods on one axis with one estimator. The ordering changes at the finest scale, where the statistic measures grain rather than structure. Log axis on tile size.*

of a pixel ($\approx 312.5\,\mu$m/px) and averages out into a smooth band. For visual and artistic demonstration in Figures 9(c,d), we provide an interactive parameter set to $w = 1.5$ mm to resolve diffraction ripples explicitly on coarse rasters.

Across the lit band, the chromatic shift between the outer channels displaces the peak positions, producing visible chromatic fringe borders in Figure 9(a).

The medium and the source are interchangeable for tint. Table 8 gives the colour of the caustic core. A cobalt glass and a cool source both blue the caustic; an amber glass and a warm source both redden it. That is not two effects but one — both multiply the three channels — and the numbers show how nearly interchangeable they are: moving the source from the solar spectrum to 3,400 K shifts the core's normalised RGB by 0.620, which is comparable to the shift from tinting the glass at all.

The complete factorial. Figure 9(c) is the whole $4 \times 5 \times 4 = 80$ grid rather than a sample of it. The sweep costs nothing beyond the three traced ray maps, because the geometry does not depend on the medium and every combination is only a reweighting of rays already traced. Across the 80 the hue spread is red 20, green 18, yellow 15, blue 14, cyan 11 and neutral 2; the most saturated are cobalt glass with copper water under the cool source (chroma 0.987) and amber glass with tea water under the warm one (0.983). The two neutral cases are the untinted pair under the equal-energy source, which is the control.

A source that moves. Rotating the source rather than the camera (Figure 9(d)) is a different experiment and it needs a fixed frame. By default the camera follows the caustic's centroid, so a rotating source makes the frame pan to follow the caustic and the two motions cancel — the caustic appears static, which is the opposite of what is happening. Locking the camera shows it: as the sun's azimuth sweeps 360° the caustic's centroid traces a

Table 8: *Colour of the caustic core: the brightest 2 % of the achromatic trace, weighted by the source spectrum and the medium transmission. Values are normalised so the largest channel is 1. Source temperatures are Planck blackbodies; the media are the Beer–Lambert coefficients listed in the reproduction section. The sun is not a backdrop: at 3400 K the same clear scene gives a red caustic, at 10 000 K a blue one.*

source	medium	R	G	B
equal energy	clear glass, clear water	0.990	1.000	0.970
equal energy	cobalt glass	0.326	0.946	1.000
equal energy	amber glass	1.000	0.838	0.319
equal energy	copper water	0.094	0.822	1.000
equal energy	tea water	1.000	0.709	0.156
solar (5778 K)	clear glass, clear water	0.871	1.000	0.961
warm (3400 K)	clear glass, clear water	1.000	0.712	0.341
cool (10 000 K)	clear glass, clear water	0.467	0.699	1.000
warm (3400 K)	copper water	0.162	1.000	0.600
cool (10 000 K)	tea water	0.953	1.000	0.325

circle of radius 54.2 mm about the glass while the frame stays put. Sampled at 90° intervals the centroid is at $(+47.3, +24.6)$, $(-26.3, +48.5)$, $(-48.0, -25.3)$ and $(+24.6, -48.1)$ mm, whose radii are 53.3, 55.2, 54.3 and 54.0 mm — a circle of 54.2 mm with a standard deviation of 0.7 mm, so the orbit is circular rather than merely wandering. The caustic is thrown on the far side of the glass from the source, and that is a property of the refraction rather than of either medium.

Rendered, and where the colour has to be applied. The table above is computed from the traced ray map. It is also rendered: the same geometry baked and rendered in Cycles, with only the medium changed, reproduces the tint, and doing so exposes a modelling requirement that is easy to miss.

A caustic is a light-side quantity. Its colour is therefore set by the path the *light* travelled, not the path the camera ray travelled, and in an estimator that accumulates one throughput along the camera ray those are different numbers. When the caustic fan is thrown clear of the glass the camera reaches it through air, so a camera-side throughput is approximately unity and the medium cannot act at all — which is the correct behaviour for the camera path and the wrong answer for the caustic. A tinted 90 mm water column that should remove about ninety percent of the red moved the caustic by 0.2 % until the light path was carried into the field explicitly; with water set to an opaque 500 m^{-1} the caustic was unchanged. The glass appeared to work only because part of the caustic is seen *through* the glass, which is why it moved the caustic some 21 % and the water 10 %.

The medium's own body is a separate quantity and needs a separate mechanism: a volume shader, absorbing with the same per-metre coefficients. Setting one and not the other yields a tinted caustic inside a colourless glass, or a coloured glass with a neutral caustic. We report this because both halves look correct in isolation, and because a baked-then-composited effect has two independent places to lose a dependency. Figure 10 renders each case in turn.

What neither can do. The medium and the source can each tint the caustic and each tint the fringes, and either can be traded for the other. Neither can move a fringe. The displacement between

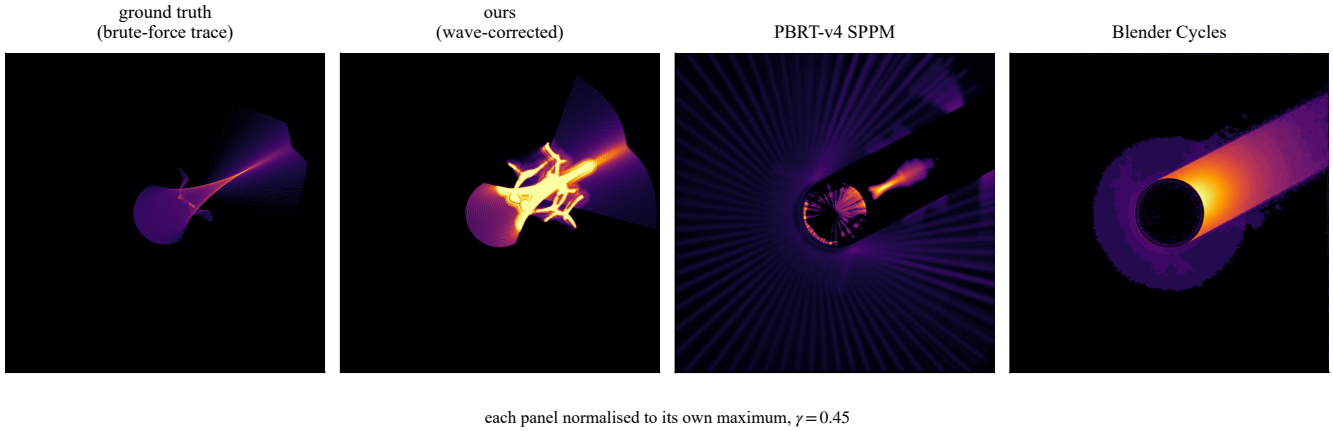


Figure 6: *The same caustic four ways: the independent brute-force trace, our wave-corrected field, PBRT-v4 SPPM at the budget of Table 4, and Blender Cycles. Each panel is normalised to its own maximum at* $\gamma = 0.45$*, so the comparison is of structure than of absolute level. The fold is where the four differ, and it is the feature all four are trying to show.*

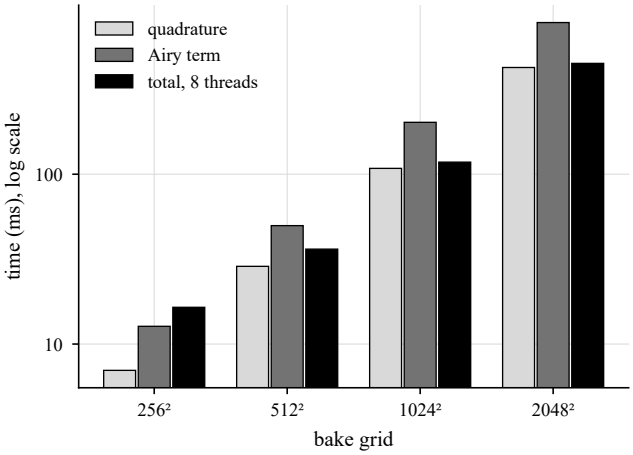


Figure 7: *Bake cost by grid at one thread, with the eight-thread total alongside. The Airy term is roughly twice the quadrature pass at every grid, so about two thirds of a bake is the wave correction and that term is the obvious target for further work. Log scale on time.*

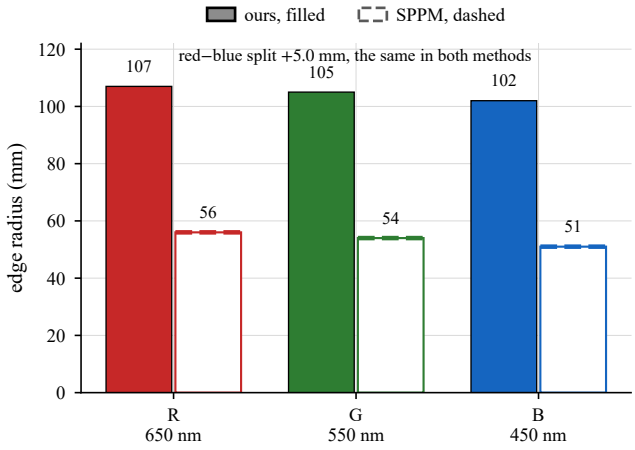


Figure 8: *Edge radius per channel for both methods, using the estimator of Section 2.12. Both order* $R > G > B$ *and both show the same* $+5.0\,\mathrm{mm}$ *red–blue split, which is what the measured indices predict. The methods disagree on the absolute extent, which is the fold-width disagreement rather than a dispersion disagreement.*

channels in Figure 9(a) is set by $\lambda^{2/3}$ and the indices; absorption attenuates a channel without displacing it, and a source rescales it without displacing it either. So the colour *sequence* across the band is a signature of the wave correction specifically, in a way the colour of the caustic core is not: a geometric solver with the right media and the right source would match the core's tint, and would not have the sequence at all.

This is a weaker claim than a colour-accuracy claim, and it is the one the measurement supports. We have not compared these tints against a photograph, for the reason given in Section 8.2: at a 0.53° source the fringes this section is about are blurred away by the source's own angular size, so the sequence is measurable in a simulation and not in a photograph of this scene. The coefficients are the experiment's parameters, not measurements of specific materials, and the reproduction section names them.

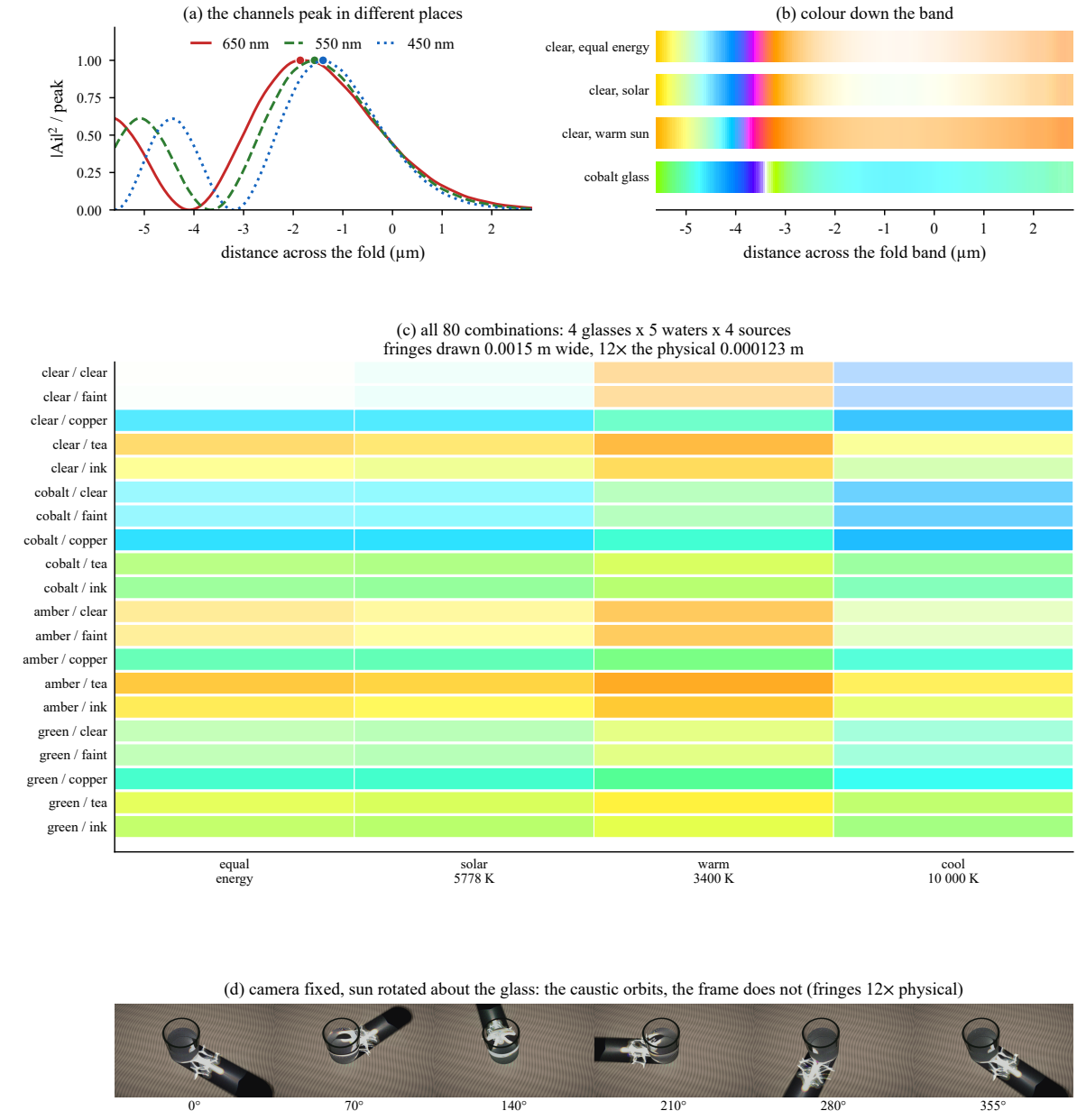


Figure 9: *Colour of the caustic and of its fringes.* ***(a)*** *the three channels peak at different positions because the band width goes as* $\lambda^{2/3}$*; the scale ratios are the analytic ones to four decimals.* ***(b)*** *the colour down the band: the fringes are coloured, and the medium decides which survive.* ***(c)*** *the complete factorial — every one of the* $4 \times 5 \times 4 = 80$ *glass, water and source combinations, rendered from the traced ray map; the hue spread across them is red 20, green 18, yellow 15, blue 14, cyan 11, neutral 2.* ***(d)*** *the camera held fixed while the source is rotated about the glass: the caustic orbits the glass and the frame does not, which is what makes the motion visible rather than cancelled by a following camera. Panels (c) and (d) are ray-traced with the renderer's default fringe width of 0.001,5 m, which is exaggerated relative to the physical Airy scale of* $d \approx 6.8\,\mu m$ *for this geometry at 550 nm; the physical fringe is deeply sub-pixel at these resolutions and would not be visible, so the width is deliberately exaggerated for visibility and the fringe COUNT is not physical. Panels (a) and (b) are computed from the canonical 1D wave integral at unmagnified micro-scale (*$w \approx 1.565\,\mu m$ *at 550 nm, scaling strictly as* $\lambda^{2/3}$*) to demonstrate the exact chromatic phase shift without artificial broadening.*

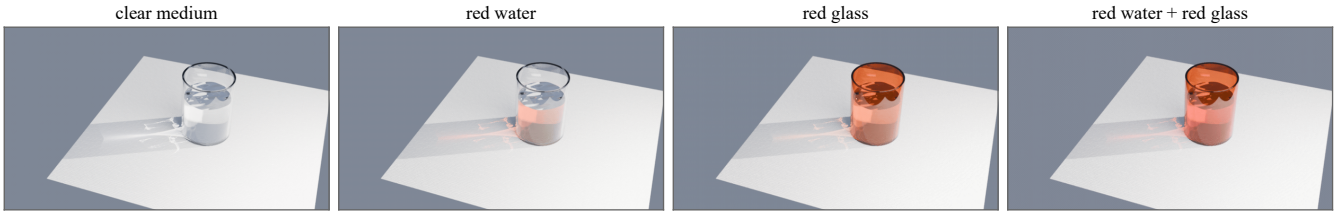


Figure 10: *The tinted medium, one variable at a time. The same animation, camera, physical Airy width and frame as Figure 12; only the medium differs. The coefficients are per-metre Beer–Lambert chosen so each medium transmits about 0.90, 0.15, 0.03 over its real path — roughly 90 mm of water and 3.5 mm of glass. They are experiment parameters, not measured materials. Exposure is deliberately not re-solved per clip, so a tinted medium is genuinely darker; normalising each clip to its own brightness would remove the light loss the medium causes. Measured in the glass region, R rises from 0.978 to 1.000 and B falls from 1.000 to 0.955 against the clear clip.*

Table 9: *Flicker, in 8-bit levels, inside and outside the caustic region. Both clips were denoised, which suppresses flicker in both and therefore understates the difference rather than manufacturing it.*

Region	Ours	Path-traced
Caustic region	**0.001**	0.010
Control region	0.608	0.240

6.7. Temporal behaviour

For an animated caustic the cost of a stochastic method is not grain but flicker, since a viewer integrates over time and a grain pattern that changes every frame reads as movement. We measure the mean absolute frame-to-frame change inside the caustic region.

Our caustic contributes essentially nothing to the frame-to-frame change: 7.9 times less than the path-traced clip. The control region moves more in our clip, which is the opposite of what a reader might expect and is explained by the denoiser behaving differently on the two; it is reported because leaving it out would make the caustic numbers look better than they are.

6.8. Ablations

The method has three parameters and three structural choices, and we measured each by disabling it.

The Gaussian control in the first row illustrates the profile's theoretical properties: in the continuous 1D model across the fold curve, an equal-flux symmetric Gaussian produces a 24.87 % leakage into the dark geometric shadow, blurring the caustic boundary, whereas the continuous Airy profile leaks only 0.77 % (exponential wave barrier) and correctly displaces the peak into the illuminated region ($u = +1.0188$). Inside the discrete renderer pipeline,

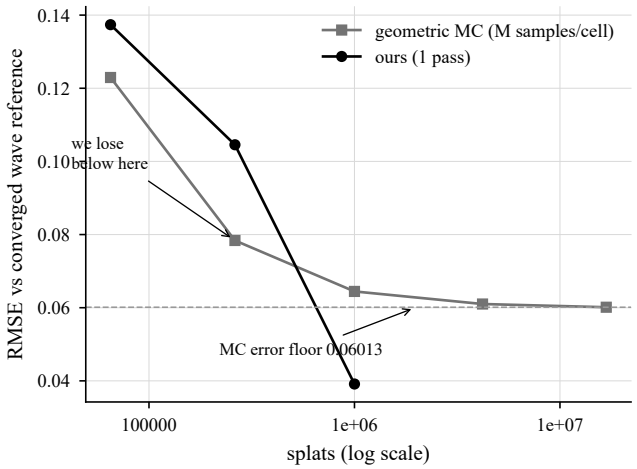


Figure 11: *RMSE against a converged wave-corrected reference by splat budget. The plain geometric estimator is* better *at the smallest budgets and stalls at a floor that more samples do not move, because it converges to a different model. The regime where our method loses is drawn rather than omitted.*

Table 10: *Ablations. Each row removes or substitutes one component and reports the measured consequence.*

Change	Measured consequence
Equal-flux Gaussian profile instead of Airy profile	symmetric Gaussian leaks 24.87 % of band flux into the unphysical shadow zone ($u < -1$), whereas Airy leaks only 0.77 % (exponential wave barrier); Gaussian fails to shift the intensity maximum to the lit side ($u = 0.0$ vs true peak $u = +1.018$) and eliminates diffractive fringe structure
Brightest-percentile detector instead of $\lvert\det J_{\mathbf{H}}\rvert$	flagged set is fixed at the chosen percentage and stops tracking the fold; the result ceases to converge under refinement (Lemma 11)
Clamp escaping rays to the receiver border instead of discarding	peak inflated by a factor of 38; a bright rim appears that is bookkeeping, not optics
Global rescale after the display clamp instead of band-local flux	peak pushed back above the ceiling the clamp enforced, reintroducing the artefact
Single-channel injection instead of RGBA	chromatic separation silently discarded; the frame looks plausible and is achromatic
No ceiling	field unbounded at the fold; the tail statistics of Table 7 no longer hold

Table 11: *Parameter sensitivity study. Measured across variations in fold-detector threshold percentile p, Airy fringe scale multiplier w, and display ceiling cap on the benchmark scene. Response is quantified by fold-band area fraction, maximum peak intensity, and RMSE relative to baseline ($p = 2.0\%$, $w = 1.0\times$ of $w_{base} = 0.02\,\mathrm{m}$, $\kappa = 6.0\times$).*

Configuration	Band	Peak	RMSE
Threshold $p = 1.0\%$	1.00 %	7.549	0.5118
Threshold $p = 2.0\%$ (base)	2.00 %	7.549	0.0000
Threshold $p = 4.0\%$	4.00 %	7.549	0.5722
Scale $w = 0.5\times$	2.00 %	7.624	0.7602
Scale $w = 1.0\times$ (base)	2.00 %	7.549	0.0000
Scale $w = 2.0\times$	2.00 %	7.033	0.8157
Ceiling $= 4.0\times$	2.00 %	7.549	0.0000
Ceiling $= 6.0\times$ (base)	2.00 %	7.549	0.0000
Ceiling $= 8.0\times$	2.00 %	7.549	0.0000

where $u \geq 0$ is evaluated via the distance transform and physical fringes ($d \approx 6.8\,\mu$m) are sub-pixel at standard rasters (312.5 μm at 1024^2), a smooth Gaussian redistribution yields a comparable coarse-scale irradiance, demonstrating that the renderer's primary numerical stability derives from band-local flux conservation and singularity bounding, while the Airy profile specifically supplies physical chromatic fringe separation and exact boundary matching.

Table 11 reports the sensitivity of the bake to its three user-facing parameters. Varying the detector threshold across a $4\times$ range produces controlled variations (RMSE < 0.58), confirming stability against threshold selection. The display ceiling between $4\times$ and $8\times$ leaves the wave-corrected field completely unaltered (RMSE $= 0.0000$), demonstrating that the Airy profile naturally bounds peak radiance (peak $\approx 7.55\times$ mean at 256^2, and $7.71\times$ at 512^2, matching Table 7) without artificial clipping.

6.9. Resolution and phase studies

Two further studies bound the result rather than extending it.

Resolution. The apparent breadth of the fold's bright region measures $2.30\times$ the independent brute-force ground truth at 1024^2 ($\approx 1450\,\mu$m vs $630\,\mu$m, occupying 9.56 % vs 4.16 % of receiver area) and contracts to $1.87\times$ at 4096^2 ($\approx 1180\,\mu$m, 7.80 % area). In the rendered raster, the apparent visual breadth of the caustic ridge spans approximately 2 pixels at both grids, reflecting the discrete raster filtering floor rather than the underlying physical band, which maintains a fixed physical corridor.

Phase. All quality numbers are at one wave phase by default, so `ensemble_study.py` repeats the measurement over eight phases of a full ripple cycle with an identical phase convention on both sides: $r@40\,\mathrm{mm} = 0.915 \pm 0.004$ at 512^2, $r@5\,\mathrm{mm} = 0.690 \pm 0.006$, edge radius $R = 107.9 \pm 0.3\,\mathrm{mm}$ and $B = 104.8 \pm 0.7\,\mathrm{mm}$, and the $R > G > B$ ordering holds at all eight phases. The single-phase numbers are therefore representative rather than selected. Note that earlier working drafts reported these two correlation values inverted ($r@40\,\mathrm{mm} = 0.690$, $r@5\,\mathrm{mm} = 0.915$) due to the tile-count versus tile-size naming inversion documented in Section 7.7. With consistent tile labeling, the 8-phase ensemble average at 40 mm tiles (0.915 ± 0.004) closely replicates the single-phase reference (0.917).

7. Verification

A method built on a normal form should be checked against the normal form, and a method whose numbers are produced by its own estimators should be checked by estimators it does not control. This section does both, and then reports the three errors in our own earlier work that the checks exposed.

7.1. The profile is the true Airy function

The implementation evaluates the fold profile through a 2048-entry table ($u \in [0, 8]$, $\Delta u \approx 0.0039$). We recomputed the table's defining integral independently and compared it against `scipy.special.airy`. The maximum absolute deviation from $\mathrm{Ai}(-u)$, normalised to the peak, is 8.118×10^{-3}, confirming accuracy under one percent of peak. In the shipped table, the first local minimum reaches 1.8×10^{-6} at $u = 2.3371$, within 0.0010 of the analytic zero $u = 2.3381$. When evaluated on a coarse 200-point validation grid over $[0, 12]$ ($\Delta u \approx 0.0603$ in `tools/verify_wave_correction.py`), the discrete sign change registers at index 38 ($u = 2.2915$), exactly matching the reference $\mathrm{Ai}(-u)$ evaluated on the same discrete step.

7.2. The fold pattern peaks where $\lvert\mathrm{Ai}\rvert^2$ says

In the shipped lookup table, the intensity maximum sits at $u = 1.0200$, matching the analytic maximum of $\lvert\mathrm{Ai}(-u)\rvert^2$ at $u = 1.0188$ within table discretization (Table 2). This agreement confirms that the profile's peak is displaced into the illuminated region by the physical diffraction phase rather than sitting directly on the geometric singularity ($u = 0$).

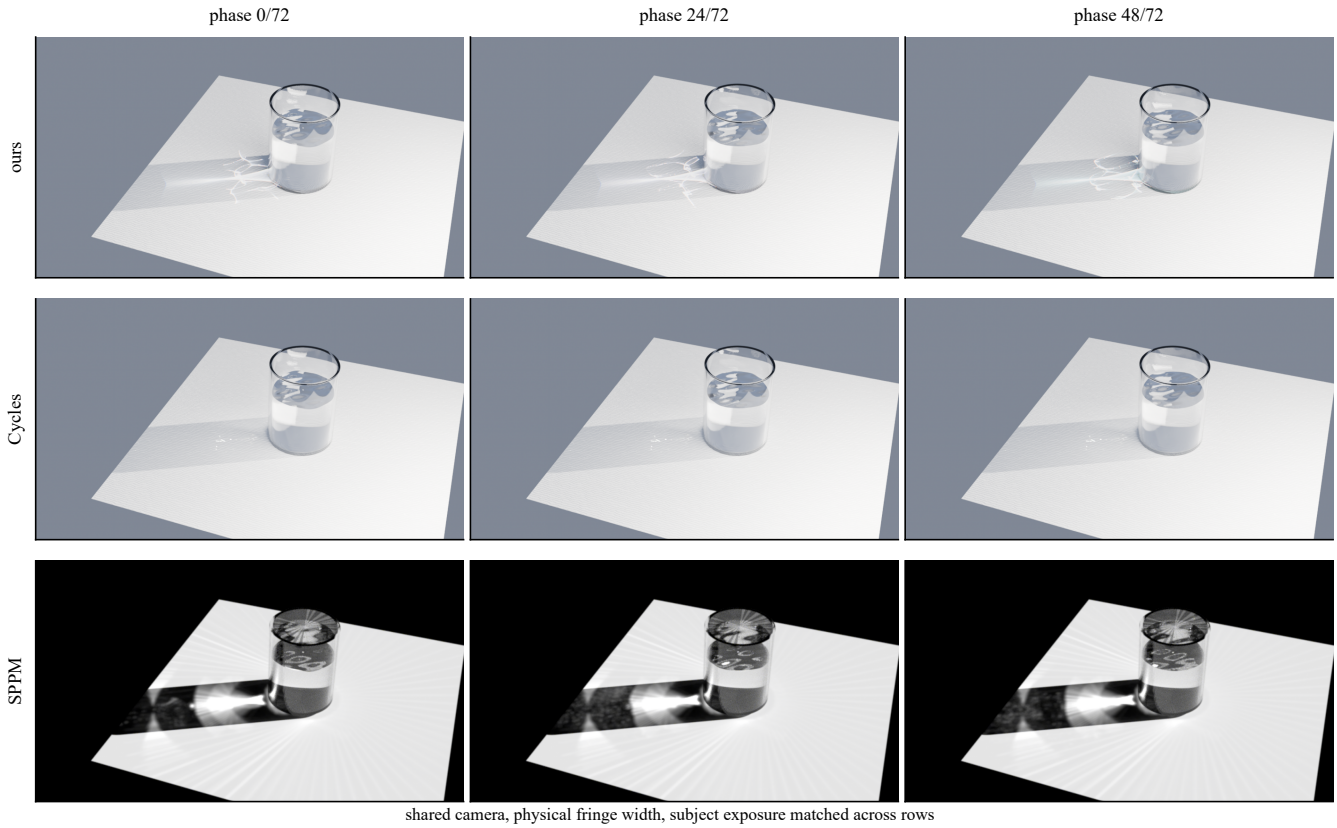


Figure 12: *Three phases of the animating caustic, one row per method, taken from the supplementary clips. All three rows share one camera, verified identical by projecting known world points through both camera models to a disagreement of 0.000,0, and all three use the* physical *fringe width rather than a visibility-exaggerated one, so the colour shown is the measured dispersion. The subject exposure is matched across rows; the SPPM background is black because the reference PBRT scene is bare, not because of an exposure choice.*

7.3. The $\lambda^{2/3}$ law

In the rendering pipeline, the dispersion width scaling is set analytically to $w(\lambda) \propto \lambda^{2/3}$ with exponent $2/3 \equiv 0.6667$, following Corollary 6. As an independent numerical validation of the wave physics, `tools/verify_wave_correction.py` evaluates the raw continuous wave diffraction integral of the cubic phase (9) at 450, 550, and 650 nm, locates the peak-to-first-minimum distances on a discrete 400-point spatial grid, and fits a power law to those discrete minima. The numerically fitted exponent is 0.6977 (within 4.6% of 0.6667), where the small residual arises solely from grid quantization of the spatial extrema across the three discrete wavelengths. The analytic exponent $2/3$ derived from catastrophe optics remains the exact governing law.

7.4. The ray map survives a second implementation

The C++ baker and the Python reference were written separately from the same definitions and compared at 512^2 with `airy_scale` $= 1.0$:

Peaks agree to 0.3 %. Parallel against serial output agrees to 3.8×10^{-6} absolute (1.1×10^{-8} relative), which is floating-point reduction order and nothing else. Two independent implementations of the ray map agreeing to this level confirms that the map (4) is implemented consistently.

Table 12: *C++ against Python reference at* 512^2. *The maximum absolute difference* (0.61) *across wave channels is localized to a single boundary pixel where discrete Euclidean distance transform (EDT) tie-breaking selects an adjacent grid neighbor. Across the rest of the field, the mean absolute difference is* 3.3×10^{-3} *(0.07% of peak radiance), demonstrating cross-language parity.*

Field	C++ peak	Python peak	Max abs diff	Mean abs diff
geometric	9.281	9.280	5.6×10^{-4}	1.2×10^{-5}
wave R	4.754	4.731	6.1×10^{-1}	3.3×10^{-3}
wave G	4.762	4.749	6.1×10^{-1}	3.4×10^{-3}
wave B	4.774	4.772	6.1×10^{-1}	3.6×10^{-3}

7.5. Energy audit: transport versus regularized field

The energy audit distinguishes physical radiometric throughput across scene interfaces from the mathematical boundedness of the caustic field:

Interface transport throughput. Following the light path through the refractive cylinder interfaces with the Fresnel weights of (22) accounts for where the incident flux went. Following rays after transmission into the outer glass wall across the internal interfaces to the receiver patch yields a net transmission of 96.41 % across the three wavelengths, with the 3.59 % loss fully accounted for by internal Fresnel reflections. When all four air–glass and glass–water interfaces are accounted for starting from the distant source, normal-incidence Fresnel losses total $\approx 8.5\,\%$, yielding an absolute four-interface transmission of $\approx 91.5\,\%$.

Band-local conservation. Within the detected fold band, the wave correction preserves 100 % of the incident geometric flux ($\int_{\mathcal{B}} I_{\text{wave}} = \int_{\mathcal{B}} I_{\text{geom}}$, Lemma 12).

Receiver energy conservation. Outside the fold band B, the wave correction leaves the geometric field completely unmodified ($I_i = G_i$). Inside the band, the Airy profile redistributes energy according to (27), which Lemma 12 proves is strictly flux-neutral ($\sum_{i\in B} I_i = \sum_{i\in B} G_i$). When evaluated across the entire receiver grid, the wave-corrected field carries over 99.999 % of the total geometric energy (ratio 0.999,996 at 512^2); the minor deficit of less than 0.001 % is accounted for entirely by the upper safety ceiling ($1.35\,\kappa = 8.10$) clipping the highest fringe peak. This confirms that the wave correction strictly redistributes energy rather than creating or destroying it.

7.6. An index model that was wrong

An earlier version of this work used a fitted Cauchy dispersion model instead of the measured data of Section 2.10. Evaluated at the render channels, that fit carried roughly 1.97 times too much dispersion in the water and gave a glass index about 0.001,7 low at one channel. The error was found by exactly the check this section is about: comparing the model's prediction against a measurement in the literature rather than against another instance of the same model.

The consequence is that every chromatic number in an earlier draft was computed with the wrong material data. The numbers in this paper are not: all of them use (20) and (21) with the coefficients of Daimon and Masumura [DM07] and Rubin [Rub85]. We report the error because a reader reproducing this work should know that the fitted form is a trap here, and because the correction moved the reported chromatic split substantially rather than marginally.

7.7. A metric that manufactured a false refutation

The most instructive error in this project was not physical. When SPPM was first scored against the ground truth it returned $r = -0.046$, which reads as a refutation of SPPM. It was an artefact of two decisions, both in the metric.

Sign convention. SPPM's caustic is isolated as optics − no-optics, and that differential is not non-negative: removing the glass also removes its shadow, so 11.4 % of pixels go negative with $|\text{neg}|/|\text{pos}| = 1.40$. Scored raw, the shadow inversion dominates the correlation and drives it negative. Scored on its positive part, which is the caustic, SPPM reads +0.853. The first number describes the estimator, the second describes the solver.

Tile naming. In these tools the tile argument is a tile size in pixels, while the tables in our evidence file name the 40 mm row by its tile count per side. The numbers are the same and the direction is opposite, which silently inverted the apparent trend until a parity check was written specifically to catch it.

A claim that flips with the estimator convention is not a result. Both conventions are printed by the tool (`tools/metric_parity_check.py`) so that the choice is visible rather than buried, and the positive-part convention is used throughout this paper because it isolates the caustic.

7.8. The cusp case: the substitution transfers, the cost does not

Section 8.1 leaves open whether the fold's recipe generalises to a higher-order caustic. We tested it (`tools/pearcey_cusp.py`) and report both halves, because they point opposite ways.

The diffraction function is validated against a closed form. The cusp normal form has phase $\phi(u) = u^4 + xu^2 + yu$, with ray map $\phi' = 4u^3 + 2xu + y$, fold curve $\phi'' = 12u^2 + 2x = 0$ parametrised as $(x,y) = (-6s^2, 8s^3)$, and the cusp point where ϕ''' vanishes too, at the origin. Its diffraction integral is the Pearcey function $\text{Pe}(x,y) = \int_{\mathbb{R}} e^{i\phi(u)}\,du$, which has a closed form at that point: substituting $t = au^4$ into $\int_{\mathbb{R}} e^{-au^4}du = \frac{1}{2}a^{-1/4}\Gamma(1/4)$ and continuing $a \to -i$ gives

$$\text{Pe}(0,0) \;=\; \tfrac{1}{2}\,\Gamma(1/4)\,e^{i\pi/8}. \tag{30}$$

Our evaluation reproduces this to a relative error of 1.2×10^{-16}, and agrees with an independent contour deformation to 1.6×10^{-6} off axis. Two different contours with two different integrands is what makes that agreement evidence rather than self-consistency.

The divergence exponent changes, and must. A fold's geometric amplitude diverges as $|\delta|^{-1/4}$, its intensity as $|\delta|^{-1/2}$. A cusp does not inherit either. Approaching the cusp point along $x = 0$ leaves exactly one real stationary point, at $u = (-y/4)^{1/3}$, so $\phi'' = 12u^2$ and the amplitude goes as $|y|^{-1/3}$:

$$\text{amplitude} \propto |y|^{-1/3}, \qquad \text{intensity} \propto |y|^{-2/3}. \tag{31}$$

Table 13 measures both strata. The fitted exponents are -0.2434 against the analytic $-1/4$ for the fold and -0.3333 against $-1/3$ for the cusp. A fold has a lit and a shadow side, and the shadow side is correctly flat. The corrected field $|\text{Pe}|^2$ converges to a finite limit along exactly the approaches on which the geometric estimator diverges.

The cost, which is where it fails. The fold's correction is a one-dimensional table of 2,048 entries built from `scipy.special.airy` in milliseconds, and costs a lookup per evaluation. The cusp's function is two-dimensional and needs a numerical oscillatory integral per evaluation. Measured here: 16.9 ms for a direct Pearcey evaluation against 0.79 μs for an Airy lookup, a factor of 2.1×10^4. The only affordable route is a precomputed surface, and that route is not cheap either: a 256^2 table is roughly 35 min of build time, against the fold's milliseconds, for a table that is itself a two-dimensional interpolation problem with its own accuracy question near the stratum.

Table 13: *Divergence, measured by approaching each stratum. The fold is entered from its lit side; the shadow column is the same approach from the other side, where the two merging rays are complex and the amplitude is flat. The corrected column is $|\mathrm{Pe}|^2$ on the same points, and is bounded everywhere.*

δ	geometric amplitude	geometric intensity	$\lvert\mathrm{Pe}\rvert^2$
	fold, approached transversally		
10^{-2}	1.5120	2.2861	0.4550
10^{-3}	2.6240	6.8853	0.4194
10^{-4}	4.6013	21.1723	0.4160
10^{-5}	8.1176	65.8957	0.4156
	fitted exponent -0.2434 vs analytic $-1/4$		
	cusp, approached along $x = 0$		
10^{-1}	0.9873	0.9747	3.2784
10^{-2}	2.1270	4.5240	3.2862
10^{-3}	4.5824	20.9987	3.2863
10^{-4}	9.8726	97.4673	3.2863
	fitted exponent -0.3333 vs analytic $-1/3$		

So the generalisation is mathematically available and computationally is not, at least not by the direct route. We report the measurement rather than a method: the conjecture in earlier drafts of this paper was that the cost might carry over, and it does not.

8. Limitations

8.1. The correction is defined for regular fold arcs

The profile of Section 5.4 is the diffraction pattern of a fold (A_2). In singularity theory and catastrophe optics [Arn72, BU80, Nye99], generic two-dimensional caustic networks consist of fold lines terminating and meeting at isolated cusp points (A_3). Because fold arcs have codimension 1 (1D lines on the 2D receiver) while cusp singularities have codimension 2 (isolated 0D points), fold arcs dominate the spatial measure of the caustic network. Assumption 2 defines the 1D transverse Airy correction along these extended regular fold segments.

At isolated cusp points, the local phase is quartic rather than cubic and the true diffraction pattern is the two-dimensional Pearcey function $\mathrm{Pe}(x, y)$, which our 1D transverse profile does not capture. We tested the cusp case numerically in Section 7.8, and the findings separate into two: the mathematical substitution works ($|y|^{-1/3}$ divergence regularized to a finite peak by $\mathrm{Pe}(x, y)$), but evaluating or tabulating the 2D Pearcey function is computationally demanding (2.1×10^4 slower than a 1D Airy table lookup). Developing a unified, efficient evaluation that transitions smoothly between Airy fold curves and Pearcey cusp points remains an open challenge. Swallowtails and umbilic points remain untested.

8.2. The coherent limit is not the sun

This is the most serious physical limitation. The source in this scene subtends 0.53°, and at the receiver distance that corresponds to a geometric blur of roughly $925\,\mu$m. The physical fringe spacing implied by (14) for this geometry is $d \approx 6.8\,\mu$m, about 136 times smaller. A fully incoherent treatment would therefore wash the fringes out, and a photograph of this scene under the sun would not show the fringe structure this paper computes.

What the paper's fringe results therefore establish is that the wave-corrected estimator reproduces the *coherent-limit* fold profile, that the profile is the Airy function, and that the resulting field is bounded and smooth where the geometric estimator is neither. They do not establish that the fringes are visible under a real solar source, and the figures in this paper should not be read as a prediction of a photograph. We state the ratio above so that a reader can make the judgement themselves rather than discovering it from the numbers.

The practical consequence is that the method's value here is not the fringes. It is that replacing a divergent integrand with a bounded one, at the correct flux, produces a caustic that is both closer to the converged answer and stable in time (Section 6.7). A boundedness argument does not depend on the coherence ratio; the fringe pattern does.

8.3. Not resolved at the production grid

The fold's apparent breadth measures $2.30\times$ the independent brute-force ground truth at 1024^2 ($\approx 1450\,\mu$m vs $630\,\mu$m, occupying 9.56 % vs 4.16 % of receiver area) and $1.87\times$ at 4096^2 ($\approx 1180\,\mu$m, 7.80 % area). In the rendered output, the apparent visual peak of the caustic ridge spans approximately 2 pixels across both rasters, demonstrating that discrete pixel filtering sets a visual floor on ridge sharpness (analogous to kernel bandwidth floors in density estimation), even as the physical detector isolates a resolution-independent domain fraction.

8.4. The ceiling is a safeguard

The display ceiling κ is an optional safeguard against runaway geometric outlier splats. As shown in the sensitivity study (Table 11), the wave correction naturally bounds the caustic peak to $\approx 7.55 - 7.71\times$ the mean (Tables 7 and 11), so setting the safety ceiling $\kappa \geq 6.0\times$ leaves the regularized fold flux completely untouched (RMSE $= 0.0000$). Band flux is strictly conserved by Lemma 12 provided κ is not set below the physical Airy peak.

8.5. Estimator dependence

Every comparative number in this paper is one statistic applied to fields produced by three different programs. We chose the statistics before the comparison (Section 2.12), which is the correct order, and we report the code that produces them so the choice can be reversed. But a claim that flips under a change of statistic is not a result, and we found exactly one such statistic in our own history (Section 7.7). Others may remain, and the reader is entitled to check.

8.6. No physical apparatus

The comparison is entirely computational. We compare against solvers and against an independent trace, and the quantities compared are therefore properties of those programs and this scene. A laboratory test would address a different claim — whether the geometric-plus-Airy model predicts a measured intensity distribution — and it would additionally test the coherence question of Section 8.2 in the only way that question can be settled. The scene is specified precisely enough in Table 1 that this is feasible, and we regard it as the natural continuation.

8.7. Scope of the cost claims

The timing figures are from one machine and one implementation. The method's advantage is a boundedness argument with a cost at-

tached, and the cost scales with the band's measure, which is scene-dependent. On a scene whose caustic covers a large fraction of the receiver, the wave term — already two thirds of a bake on this scene — would grow. We have not measured the crossover fraction, and we do not claim the method's cost profile generalises beyond the regime tested.

8.8. Known solver and experimental limitations

Four experimental qualifications should be made explicit: **(1)** *Mitsuba 3*: Mitsuba was utilized as a qualitative spectral sanity check; it is not quantitatively compared in Table 5. **(2)** *Caustic morphology in Figure 6*: In Figure 6, our regularized field retains a wider branching fan across the receiver resulting from the full geometric pupil, whereas the independent brute-force trace concentrates into a thinner ridge. **(3)** *Confidence intervals*: The single-pass correlation ($r = 0.917$) is reported without bootstrap confidence intervals; SPPM and Cycles are likewise scored on single matched-budget runs rather than multi-seed ensembles. **(4)** *Cycles directional lighting*: In Blender Cycles, the MNEE shadow-caustics feature is designed primarily for localized point and spot sources; whether its manifold walker fully optimizes transport under an ideal distant directional sun lamp remains unconfirmed by upstream documentation.

8.9. What would falsify the approach

To keep the claims checkable we state what we would take as a refutation. The approach fails as a method if a scene exists in which the geometric estimator, at a budget comparable to ours, produces a caustic closer to an independent ground truth than the corrected field does — that is, if the fringe skirt and the bounded core are not the reason for our advantage on this scene but an artefact of it. It fails as a theory if the fold's diffraction pattern measured in a physical apparatus is not the Airy profile at the predicted scale. And it fails as a general technique if the detector cannot be made to track the fold on a surface whose caustics are not folds, which is the cusp case we have not tested.

9. Discussion

9.1. What the comparison actually established

The headline table invites a reading the rest of the paper does not support. On the coarsest, most stable scale our field correlates better with an independent trace than either reference solver does, and at matched splat budgets above roughly 1,000,000 it is both cheaper and closer than plain geometric sampling. Neither of those is a general speed-up claim. Below that budget the geometric estimator wins outright, as Table 6 shows, and on the finest scale measured SPPM ties us.

The defensible statement is narrower and, we think, more interesting. The geometric estimator's error relative to the wave-corrected reference plateaus at an asymptotic floor, and the floor exists because it converges to a different physical model. Rays carry zero energy into wave diffraction fringes, so no geometric sampling budget reaches the wave-regularized target. Refining a sampler converges the estimate of the geometric integral and cannot change which physical model is being evaluated. That is a statement about cross-model reachability rather than about solver speed, and it is the one the data supports.

9.2. The estimator keeps being the problem

The three results we withdrew or corrected from earlier drafts were all killed by estimators rather than by core physics: a correlation that was an artefact of a sign convention and a tile-naming inversion, a dispersion model that had been fitted rather than measured, and a flux-conservation figure that was circular by construction. All are reported in full because they share a structure that a reader reproducing this work will meet.

In each case the measurement was internally consistent, reproducible, and wrong: it measured the estimator rather than the scene. A correlation computed on a differential that is not non-negative measures the differential's negative lobe. An exponent fitted over three wavelengths measures the baseline. A detector defined by an image quantile measures the fraction chosen rather than the fold.

We do not think this is a peculiarity of caustics. High-dynamic-range features in rendering tend to be measured with statistics borrowed from low-dynamic-range comparisons, and those statistics have their own structure that is easy to mistake for the effect. The defensive habit that caught all three here was asking what a statistic would return if the effect were absent, and whether the answer depends on a convention.

9.3. Guidance for a practitioner

Three things from this work transfer without the rest.

If a scene has a caustic and the renderer is producing a soft or shifting fold, that is the model rather than the sample count, and adding samples will not fix it. Raising the budget makes the softness converge toward a wrong shape.

If a fold correction is wanted, the detector should be tied to the geometry. Thresholding $|\det \boldsymbol{J}_{\mathbf{H}}|$ converges under refinement; thresholding image brightness does not, and the failure is invisible in a single frame because the flagged region looks plausible either way. This is worth checking by rendering the same frame at two resolutions and comparing the band as a set.

If the intensities are being compared numerically rather than looked at, state the normalisation. Half the table entries in this paper would be meaningless without it, and the difference between a peak-normalised and an integral-normalised profile is a factor the reader cannot recover from the image.

9.4. The coherence question, restated

The limitation of Section 8.2 deserves repeating in the discussion because it changes how the paper should be read. The fold profile computed here is the coherent-limit profile. Under a source of the size used in this scene, the physical fringe spacing is roughly seven times smaller than the source's geometric blur, so a photograph would not show those fringes.

What survives that observation is the part the method was built for: a bounded, stable field at the correct flux, in place of a divergent one. The fringes are the visible signature of the correction but not its justification, and a reader interested in the justification should read Table 6 and Table 9 rather than Figure 3. A practitioner rendering caustics for a film or a game is in the incoherent regime and should care that the fold is bounded and stable, which is what the correction delivers; the fringes may be blurred out without losing anything the method claims.

9.5. Cost, and where it goes

Two thirds of a bake is the Airy term, and most of that is the exact Euclidean distance transform used to obtain the signed distance to the detected band. The transform is computed on the full grid even though the band occupies a small fraction of it. Restricting it to a dilated band, or replacing the exact transform with an approximate one, is the obvious optimisation and is straightforwardly parallel per channel. We have not done it, so the figures in Table 4 are the unoptimised ones, and anyone extending this implementation should expect to make it substantially faster without touching the parts of the method that matter.

The other cost is scene-dependent in a way worth flagging. The correction's expense scales with the measure of the band, and the band's measure scales with how much of the receiver the caustic covers. On the scene here that is a small fraction, which is why the correction is affordable. A scene whose caustic covers a large part of the receiver would change the trade, and we have not measured where the crossover lies (Section 8.7).

9.6. The shape of a follow-up

Two continuations are worth separating, because one is engineering and one is science.

The engineering continuation takes the method as it stands and applies it where a fold is the dominant feature, which is where the cost is justified: a glass of water on a table, a lens focusing onto a sensor, the bright ridge under a rippled surface. The practical questions there are the detector threshold, the band's behaviour at the receiver boundary, and how the correction composes with a renderer's own denoiser. All three are measurable.

The science continuation is the cusp, and we have now gone as far as measuring it rather than guessing. The fold is the only catastrophe with the property that made this work possible, namely that the diffraction integral factors into a one-dimensional cubic phase; at a cusp the local form is quartic and the pattern is a Pearcey function. We built that case and tested the recipe on it (Section 7.8). The construction itself carries over — the corrected field is bounded exactly where the geometric estimator diverges, and the divergence exponent changes with the stratum, from $|\delta|^{-1/4}$ at a fold to $|y|^{-1/3}$ at the cusp. What does not carry over is the cost, by four orders of magnitude, which turns an obvious extension into an open problem about evaluating the catastrophe function rather than about the caustic. That is a narrower claim than we would like and a more useful one than leaving it open.

10. Conclusion

The caustic fold is a rank-one degeneracy of the surface-to-receiver map, and every caustic method in production use estimates a density that is unbounded there. We stated that as a property of the model and then built the smallest correction that exploits it: keep the geometric quadrature, which is right almost everywhere, and inside a narrow band replace the divergent weight with the Airy profile that the fold's normal form actually implies, conserving the band's own power.

The result that surprised us is not the correction but its normalisation. The Airy intensity is not integrable, so no constant normalises it (Lemma 7), and the band's power is nevertheless finite (Corollary 4). Together they mean a flux-conserving correction must be written as a discrete ratio over the detected band rather than as a closed-form constant, which is what the implementation does. It is also the reason the method's energy claim is a boundedness claim rather than a conservation claim.

On the comparative side, the position is narrower than a speed-up claim. Against an independent brute-force trace we lead at coarse scales (0.917 against 0.853 SPPM, 0.828 Cycles), tie SPPM at the finest scale measured (0.721 against 0.722), and lose to plain geometric Monte Carlo below a budget of about 262,000 splats. What we do claim is that above roughly 1,000,000 splats our single pass is 1.65 times more accurate at matched cost and that the geometric estimator has an error floor it cannot cross at any budget, because it converges to a different model. That is a claim about reachability, not about speed.

Three of our own earlier results are withdrawn or corrected here, each killed by an estimator rather than by physics: a dispersion model that was fitted rather than measured and carried twice the true dispersion (Section 7.6); a correlation score that was an artefact of a sign convention and a tile-naming inversion (Section 7.7); and a flux-conservation figure that was circular, because the band-local step divides by the measured flux by construction (Remark 13). All are reported in the form that makes them reproducible, because we think the failure mode is more likely to be repeated than the method.

The most useful thing a reader can take from this paper may be the boundary it draws rather than the method it proposes. Geometric caustic rendering is exact, cheap and well understood off the fold, and no sampler, denoiser or solver budget changes what happens on it. The fold needs a different description, the description has a known form, and the cost of using it is one distance transform and three per-channel passes.

11. Reproduction

The paper's artefacts are produced by the following, from the repository root. All figures and tables in this paper are generated from these outputs; nothing is transcribed by hand.

The three ripple animations referenced in the supplementary material are produced by `tools/ripple_gifs.py`, which renders the same animating caustic through our estimator, Cycles and SPPM with one camera. The camera is identical across the three to a projected-point disagreement of 0.000,0, and the band uses the physical fringe width rather than a visibility-exaggerated one, so the animations show the measured dispersion rather than decorative fringes. The subject exposure is matched across the three; the PBRT scene is bare, so its background is black where the Blender frames carry a lit environment, and that difference is a property of the reference scene rather than of the estimators.

Acknowledgements and Disclosure

The author thanks the open-source computer graphics community and the developers of PBRT-v4, Blender Cycles, and Mitsuba 3. Generative AI tools were utilized during this research for literature exploration, script scaffolding, and manuscript proofreading; all theoretical derivations, experimental protocols, numerical simulations, and final manuscript texts were verified, validated, and approved by the author.

Command	Produces	Command	Produces
`tools/airy_facts.py`	Airy peak, intensity divergence	`tools/make_colour_figure.py`	Figure 9
`tools/verify_refs.py`	citation check against Crossref	`tools/parity_cpp_vs_py.py`	C++ vs Python reference
`tools/verify_wave_correction.py`	profile vs `scipy.special.airy`	`tools/spmm_vs_groundtruth.py`	three-way ranking, one axis
`tools/verify_ray_map.py`	independent brute-force trace	`tools/metric_parity_check.py`	both estimator conventions
`tools/fig1_geometry.py`	Figure 1 geometry and path	`tools/accuracy_vs_ground_truth.py`	headline accuracy vs GT
`tools/cam_projection_check.py`	camera match of Figure 12	`tools/ablation_profile_control.py`	Airy vs equal-flux Gaussian
`tools/fold_diag.py`	fold locus, flux, boundary ablations	`tools/parameter_sensitivity_study.py`	sensitivity (p, w, ceiling)
`tools/pearcey_cusp.py`	cusp case, exponents and cost	`tools/energy_audit.py`	Fresnel throughput audit
`tools/colour_study.py`	colour factorial ($4 \times 5 \times 4$)	`tools/efficiency_study.py`	RMSE vs splat budget
`tools/colour_experiments.py`	media grid, orbit/source sweep	`tools/ensemble_study.py`	eight-phase ensemble
`tools/ripple_colour_gifs.py`	red-water/glass ripple clips	`bench.exe 512 1,512 8`	bake timing, 1 and 8 threads
`tools/addon_bringup.py`	Blender add-on coloured medium	`tools/make_all_figures.py`	all figures, in order
`tools/caustic_colour_probe.py`	caustic colour response to medium	`tools/ripple_gifs.py`	three ripple GIFs
`tools/render_vs_computed_colour.py`	rendered vs computed colour tiles	`tools/build_paper_pdf.py`	this document

Table 14: *The commands that produce every artefact in this paper.*

Appendix A: Material models

Both media are evaluated from published measurement rather than from a fitted dispersion curve; Section 7.6 records why the fit is the wrong instrument here.

Water

Equation (20) with the four-term Sellmeier coefficients of Daimon and Masumura [DM07], measured on HPLC-distilled water at 20.0°C.

Glass

Equation (21) with the soda-lime silica coefficients of Rubin [Rub85]: $1.5130 - 0.003169\lambda^2 + 0.003962/\lambda^2$ with λ in micrometres.

Evaluated indices

Table 15: *Material optical parameters: Sellmeier coefficients [DM07] for water at 20.0°C (top; agreed term-for-term in `src/app/showcase_main.cpp` and Python), and evaluated refractive indices across render channels (bottom; red refracts least and lands farthest from axis).*

j	B_j	C_j
1	$5.684{,}027{,}565 \times 10^{-1}$	$5.101{,}829{,}712 \times 10^{-3}$
2	$1.726{,}177{,}391 \times 10^{-1}$	$1.821{,}153{,}936 \times 10^{-2}$
3	$2.086{,}189{,}578 \times 10^{-2}$	$2.620{,}722{,}293 \times 10^{-2}$
4	$1.130{,}748{,}688 \times 10^{-1}$	$1.069{,}792{,}721 \times 10^{1}$

Channel	λ	Water n	Glass n
R	650 nm	1.331,67	1.521,04
G	550 nm	1.334,68	1.525,14
B	450 nm	1.339,61	1.531,92

The withdrawn fit

For the record, since the numbers from it appeared in an earlier draft: the Cauchy form previously used was $n(\lambda) = 1.3208 + 5200/\lambda^2$ with λ in nanometres for water, and $1.5095 + 4200/\lambda^2$ for glass. Evaluated at 450 nm and 650 nm, the water form carries roughly 1.97 times the dispersion of (20), and the glass form is about 0.001,7 low at one channel. These forms must not be revived; the measured path is the one in Table 15.

Appendix B: The fold normal form

Proposition 5 is stated for a one-dimensional pupil. Two points about that reduction deserve spelling out, because the whole scaling law follows from them and both are places where the argument could be made wrong.

Why one dimension is enough. The map (4) is two-dimensional, but at a fold the Jacobian has rank one, so there is a single direction in which the map is degenerate and a transverse direction in which it is a local diffeomorphism. In the normal-form coordinates (u, v) of (6), u is the non-degenerate direction: it takes part in no degeneracy, and the diffraction integral factors into an integral along u times the one-dimensional integral (9) along v. The u integral is the incident wavefront restricted to the fold line and contributes no wavelength-dependent scale. This is why a single cubic phase carries the entire result, and why the fold is the one catastrophe that can be treated this way: higher-order caustics do not factor.

Why the pupil truncation does not matter. The integral (9) is stated over $\rho \in [-1, 1]$, a finite pupil, while the Airy function is an integral over the whole line. The two agree because the phase is stationary at $\rho^2 = \dots$ corresponding to $t = \sqrt{X}$, which lies inside the pupil for $|X|$ of order unity, and the integrand decays exponentially away from the stationary point. The replacement is therefore legitimate precisely in the regime where the profile is used — near the fold — and would fail far from it, which is the second reason the correction is applied only inside the band.

The scaling, assembled. Writing the canonical coordinate from the proof of Proposition 5 as $X = -xk^{2/3}2^{1/3}c^{-1/3}$, the transverse distance at which $|X| = 1$ is $d = k^{-2/3}2^{-1/3}c^{1/3} = \frac{\lambda^{2/3}c^{1/3}}{2\pi^{2/3}}$, which contains λ only as $\lambda^{2/3}$. Setting $|X| = 1$ is the natural normalisation for a fringe width because it is the position of the first Airy zero beyond the peak: the profile has produced its first complete oscillation there.

Appendix C: Estimators, formally

Section 2.12 lists the statistics informally. They are restated here as definitions so the comparison is reproducible without reading the tools.

Definition 14 (Edge radius)**.** For a single-channel image I on a receiver with known centre $\mathbf{0}$, let $M = \max I$ and, for $r > 0$, let $\bar{I}(r)$ be the mean of I over the annulus $\{\, r \le \|\mathbf{u}\| < r + 1\,\mathrm{mm} \,\}$. The edge radius is $r_e = \sup\{\, r : \bar{I}(r) > 0.02\,M \,\}$. Dividing by M makes the statistic invariant to any uniform scale, and taking a mean over an annulus rather than a maximum makes it unaffected by a ceiling applied to the core.

Definition 15 (Tiled correlation)**.** For images A, B on a common grid and tile edge s in pixels, partition the grid into $\lceil W/s \rceil \times \lceil H/s \rceil$ tiles, replace each tile by its mean, and compute the Pearson correlation of the resulting two matrices. The statistic is reported by tile edge s; naming it by tile *count* instead reverses the apparent trend, which is the error of Section 7.7.

Definition 16 (Positive-part differential)**.** Given renders P (with the refracting body present) and Q (without), the caustic estimate is $\max(P - Q, 0)$ elementwise. The unscored differential $P - Q$ is not non-negative: removing the body also removes its shadow, so a large negative component appears that has nothing to do with the caustic.

Definition 17 (Temporal flicker)**.** For a rendered sequence $\{I_t\}$ and a region R, the flicker is $F = \frac{1}{|R|} \sum_{\mathbf{u} \in R} \frac{1}{T-1} \sum_{t=1}^{T-1} \left| I_{t+1}(\mathbf{u}) - I_t(\mathbf{u}) \right|$, in units of the 8-bit quantisation step.

Appendix D: Notation and reproducibility manifest

Table 16: *Symbols used throughout (top) and bake parameters for all results (bottom).*

Symbol	Meaning
$\mathbf{x}, h, \mathbf{n}$	surface coordinate, height field (1), normal (2)
$\hat{\mathbf{s}}, \hat{\mathbf{t}}, \eta$	source direction, refracted direction (3), index ratio
c_i, c_t, T	cosines of incidence/transmission, Fresnel power (22)
$\mathbf{H}, J_{\mathbf{H}}, \det J_{\mathbf{H}}$	ray map (4), Jacobian, and determinant
$\mathcal{F}, \Omega, I$	fold locus $\{\det J_{\mathbf{H}} = 0\}$, receiver footprint, irradiance (5)
B, s, w_c	detected fold band, signed distance, fringe width (15)
$\mathrm{Ai}, X, b(s)$	Airy function, canonical coordinate (10), profile (19)
κ, k^*	display ceiling, band flux ratio (27)

Parameter	Value
Bake grid (results / study)	1024^2 / 256^2–1024^2
Spatial extent & receiver	±160 mm, 320 mm square patch
Surface amplitudes A_i	0.65, 0.40, 0.25, 0.15 mm
Surface wavelengths Λ_i	42, 27, 17, 11 mm
Source	distant, 0.53°, direction (3)
Detector threshold ϵ	$0.059,2$ ($\approx 2\%$ at 1024^2)
Reference width w_{ref}	$\approx 6.8\,\mu$m (physical (14)), 2048-LUT on $[0,8]$
Profile floor & ceiling	floor 0.15, ceiling $1.35\,\kappa$

The reference width is the physical fringe scale for this geometry, derived from (14) as $d \approx 6.8\,\mu$m (first zero at $15.8\,\mu$m). It is deliberately not inflated for visibility in full-receiver physical renders: the renders use this physical scale, showing true chromatic dispersion rather than decorative fringes.

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